

PROJECTIVE HODGE LINES AND ORDINARY IITAKA SUBADDITIVITY

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ABSTRACT. We prove the ordinary Iitaka subadditivity conjecture for surjective projective morphisms with connected fibers between smooth connected projective varieties over algebraically closed fields of characteristic zero. If F is the geometric generic fiber of $f : X \rightarrow Z$, then $\kappa(X) \geq \kappa(F) + \kappa(Z)$.

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1. INTRODUCTION

Let $f : X \rightarrow Z$ be a surjective morphism with connected fibres between smooth projective varieties. The pluricanonical forms on X measure its birational complexity. Iitaka's subadditivity problem asks whether the forms on the base and on the geometric generic fibre always contribute additively to that complexity. More precisely, the Kodaira dimension $\kappa(X)$ is the maximum dimension of the images of the complete linear systems $|mK_X|$, for positive integers m ; it is $-\infty$ if every such system is empty. If F is the geometric generic fibre of f , the conjecture $C_{n,m}$ asks for

$$\kappa(X) \geq \kappa(F) + \kappa(Z), \quad n = \dim X, \quad m = \dim Z.$$

We use $(-\infty) + a = -\infty$ and $\kappa(\text{point}) = 0$.

The problem belongs to Iitaka's theory of dimensions of divisors [19]. Its development has repeatedly connected the growth of pluricanonical systems with positivity on the base. Viehweg's weak-positivity method gives addition when the base is of general type [28]. Kawamata proved addition over curves and when the general fibre has a good minimal model [20, 21]; Kollár established the case of general-type fibres [22]. The log-general-type-fibre theorem of Kovács–Patakfalvi and its formulation by Hashizume provide the addition theorem used below [23, 18].

Birkar proved the conjecture in total dimension at most six [6]. Using canonical-bundle and nonvanishing methods, Chang extended this to total dimension at most seven under the additional assumption $\kappa(X) \geq 0$ [12, Theorem 1.8]. Other advances include Cao–Păun's theorem

for projective klt pairs over abelian varieties, Hacon–Popa–Schnell’s result for bases of maximal Albanese dimension, and Cao’s theorem for projective klt pairs over surfaces [9, 17, 8]. These results show how information about the base and positivity of direct images can control the total pluricanonical system. Tsuji’s preprint states ordinary subadditivity in Theorem 1.13, while Maehara’s preprint states a broader variation inequality in Section 7, Theorem 8 [27, 24]. Those preprint claims are not inputs to our proof. The argument here is a projective proof through a Hodge line and a canonical bundle formula.

Theorem 1.1 (Ordinary Iitaka subadditivity). *Let k be an algebraically closed field of characteristic zero, and let $f : X \rightarrow Z$ be a surjective projective morphism with connected fibres between smooth connected projective k -varieties. If F is its geometric generic fibre, then*

$$\kappa(X) \geq \kappa(F) + \kappa(Z).$$

Thus the ordinary Iitaka conjecture has a positive answer in this setting. The main step in the proof is a positivity statement for the entire highest Hodge line of a rationally polarized variation. We first describe that statement, then explain how the canonical bundle formula brings it into the subadditivity problem.

Let Y be a smooth connected projective complex variety and U a dense open subset. A rationally polarized integral pure variation $\mathbb{V}$ on U consists of a rational local system with a monodromy-invariant lattice, a Hodge filtration of fixed weight, and a rational flat polarization. Suppose its highest nonzero filtration step $F^p \mathbb{V}_{\mathcal{O}}$ is a line, so $F^{p+1} = 0$ and $\text{rank } F^p = 1$. On a simply connected open set, a flat marking identifies this line with a holomorphic map to the projective space of a fixed reference fibre. We call this the *highest-line map*. The *full period map* records every step of the Hodge filtration.

The highest line has a rational parabolic extension across a smooth normal-crossing compactification. This is the extension whose pullback, after a finite level cover and resolution making local monodromy unipotent, is the Deligne–Schmid highest line. The rational weights retain the finite parts of the original local monodromies.

Theorem 1.2 (Projective adjoint positivity). *Let Y be a smooth connected projective complex variety, let $U \subset Y$ be a dense Zariski open subset, and let $\mathbb{V}$ be a rationally polarized integral pure variation of Hodge structure on U whose entire highest nonzero Hodge filtration step is a line. Let $M \in \text{Pic}(Y) \otimes \mathbf{Q}$. Assume that, after a smooth projective birational modification of Y if necessary, there are a smooth projective alteration $\tau : \widehat{Y} \rightarrow Y$ and a rational number $c > 0$ such that the pulled-back variation has unipotent monodromy along a simple-normal-crossing boundary and*

$$\tau^* M \sim_{\mathbf{Q}} c J_{\widehat{Y}},$$

where $J_{\widehat{Y}}$ is its extended highest line. After further smooth projective birational modification, there are a surjective morphism $p : Y \rightarrow S$ with connected fibres onto a smooth projective variety and a nef rational line L on S such that

$$M \sim_{\mathbf{Q}} p^* L, \quad K_S + jL \text{ is big for all sufficiently large integers } j.$$

If S is a point, the conclusion is $M \sim_{\mathbf{Q}} 0$.

The two period maps have separate jobs. The full period image provides the projective base on which an adjoint divisor can be big. The numerical dimension of L is measured by the highest-line map, whose rank can be smaller. The argument must therefore remove the boundary using the rank of the line, without identifying it with the dimension of the full period image.

1.1. The boundary obstruction and its resolution. The algebraicity theorem of Bakker–Brunebarbe–Tsimmerman constructs the full period image as an algebraic variety [4, Theorem 1.1]. At a neat level, Brunebarbe–Cadorel’s theorem gives logarithmic general type when the full period map is generically immersive [7, Theorem 1.1]. A finite quotient returns to the original level. This introduces a second divisor to be removed: besides the boundary with nonzero monodromy logarithm, there is divisorial ramification of the finite quotient. Ordinary adjoint positivity requires control of both.

We first retain the smallest rational subvariation containing the highest line. This is the pure transcendental-part construction of Bakker–Filipazzi–Mauri–Tsimmerman [5, Definition 2.9]. Rational minimality has a useful consequence: a connected rational normal subgroup of monodromy that fixes the line pointwise must be trivial. Together with André’s monodromy normality theorem [3, §5] and the fixed-part results of Griffiths, Deligne, and Schmid [13, added note to 4.2.6 and 4.2.8(i),(ii)] [25, Theorem 7.22 and Corollary 7.23], it also detects a flat automorphism that is scalar on all highest lines and preserves one full period. Such an automorphism preserves every full period.

The local analytic step concerns a family of holomorphic maps approaching a divisor. If its limiting tangential rank equals the full rank in the interior, then its nearby image is already contained in the limiting image. At a nilpotent degeneration, the limiting line is in the kernel of the monodromy logarithm; rational minimality rules out equality of ranks. At a divisor fixed by finite inertia, the limiting lines lie in one eigenspace. The same rank principle and the fixed-part theorem would make the inertia element fix the entire full period image.

For this last contradiction we use the normalization of the *actual* period image. The finite group acts effectively on its function field. Equality of the full periods therefore forces the inertia element to be the identity. A further finite cover of the image could have a nontrivial automorphism acting trivially on every period, so connected-fibre factorization is taken only after this argument. Sections 2–4 give the rational, analytic, and finite-quotient parts of this rank loss in that order.

Once the ranks drop, elementary section counting removes the boundary and divisorial ramification. If the highest-line map has generic rank r , the ample perturbation in the interior supplies a polynomial of degree r in a large multiple of L , whereas the restriction polynomials on those divisors have degree at most $r - 1$. Effective divisors exceptional over the quotient do not change its section growth. This yields the adjoint positivity in Theorem 1.2.

1.2. From the period theorem to ordinary subadditivity. For the nonvacuous case of Theorem 1.1, put $d = \kappa(F) \geq 0$ and assume $\kappa(Z) \geq 0$. The relative Iitaka fibration factors a birational model of X through a variety Y with $\dim(Y/Z) = d$ and Kodaira-dimension-zero generic fibre over Y . The canonical bundle formula separates the singular-fibre contribution from a nef moduli divisor:

$$K_{\tilde{X}} \sim_{\mathbf{Q}} g^*(K_Y + B + M) + R, \quad \tilde{X} \xrightarrow{g} Y \xrightarrow{q} Z.$$

Here B is an effective klt boundary, and the residual divisor R has the direct-image and exceptionality properties needed to compare complete pluricanonical systems. This is the strategy of Fujino–Mori’s canonical bundle formula, with the birational discriminant and moduli theory of Ambro [15, Theorem 4.5] [1, 2].

The least-index canonical root cover of the Kodaira-zero fibre has geometric genus one. Its unique top form is therefore the entire highest line of a rational variation, even if the character of the cyclic covering group is not rational. The Hodge realization of M matches its extension on an alteration, not only its restriction to the open base. These facts place M within Theorem 1.2. The auxiliary sub-boundary in this formula can be negative on the generic fibre; the argument

uses nefness and the Hodge realization, not a semiampleness assertion requiring an effective generic boundary.

The section comparison gives

$$\kappa(X) \geq \kappa(Y, K_Y + B + M), \quad (K_Y + B + M)|_{Y_{\bar{\eta}}} \text{ big},$$

where $Y_{\bar{\eta}}$ is the geometric generic fibre of q . The period projection writes $M = p^*L$ for a morphism $p : Y \rightarrow S$. Write $\mu : \tilde{X} \rightarrow X$ for the birational modification of the original source in the relative Iitaka construction. Figure 1 separates the two maps from Y : their bases need not map to one another.

$$\begin{array}{ccccc} \tilde{X} & \xrightarrow{g} & Y & \xrightarrow{p} & S \\ & \searrow f\mu & \downarrow q & & \\ & & Z & & \end{array}$$

FIGURE 1. The relative Iitaka map g has Kodaira-dimension-zero generic fibre. The original base map q and the independent period map p start from the same variety Y . The divisor $M = p^*L$ comes from S , while the desired addition inequality is measured over Z .

A product-map argument first gives log pluricanonical nonvanishing on a general fibre of p . Fujino's twisted weak positivity then turns the big divisor $K_S + jL - H$, for an ample H and large j , into an effective divisor linearly equivalent to $K_Y + B + jM - p^*H$ [14, Theorem 1.1]. Interpolating this divisor with an effective klt log canonical class reduces the desired inequality to the established addition theorem for log-general-type fibres [18, Theorem 2.11]. Scalar extension of each pluricanonical space transfers the result from $\mathbf{C}$ to any algebraically closed field of characteristic zero.

There are two further proofs of the Hodge-line boundary estimate in the projective setting. An integral tensor replacement makes monodromy faithful on the moving highest line; projection to a fixed quotient by a monodromy logarithm then produces transverse loops on which the line is constant. A second calculation obtains the restriction-intersection bound directly from Hodge curvature: smooth Thom forms concentrate on a boundary divisor, and a product Poincaré estimate controls their limits at crossings. These two calculations share the integral tensor replacement and the transverse-loop criterion; the second supplies a different proof of the boundary restriction-intersection step. These arguments retain the same whole-top-line hypothesis but isolate different mechanisms behind the boundary estimate. In the main route, rational minimality and the tangential rank principle treat the finite quotient on the actual period image directly.

1.3. Organization. Section 2 constructs the rationally minimal variation and proves the fixed-period criterion. Section 3 compares numerical rank with boundary limits and proves the tangential rank principle. Section 4 applies these results to boundary monodromy and divisorial ramification. Section 5 constructs the actual period quotient and completes the proof of adjoint positivity. Section 6 gives the canonical bundle data, the least-index root cover, and the section-growth comparison. Section 7 proves ordinary subadditivity and gives the scalar-extension argument in characteristic zero. Section 8 proves the integral tensor replacement and the alternative adjoint argument by transverse loops. Section 9 proves the restriction-intersection estimate using Thom forms, including the crossing calculation.

1.4. Conventions. An algebraic fibre space is a surjective projective morphism with connected fibres. All varieties in the Hodge-theoretic argument are over $\mathbf{C}$. Divisors and line-bundle identities are rational unless otherwise specified, and section comparisons are taken in degrees clearing all denominators. We write $A \sim_{\mathbf{Q}} B$ when a positive integral multiple of $A - B$ is principal. An effective rational class means a class rationally linearly equivalent to an effective divisor. The positive and negative parts of a rational divisor have disjoint support. Numerical dimension is used only for nef divisors and is defined in Section 3.

2. RATIONAL MINIMALITY AND FIXED PERIODS

Our objective is to recognize when a symmetry that fixes all highest lines must fix the full periods. We first discard the rational subvariations that do not contribute to the highest line. Rationality is essential: the invariant spaces of rational normal monodromy subgroups will then be Hodge substructures.

Let U be a smooth connected complex algebraic variety and let $V = (V_{\mathbf{Z}}, F^{\bullet}V_{\mathbf{O}}, Q)$ be a polarized integral pure variation of Hodge structure of weight w . Integral means that the underlying rational local system has a monodromy-invariant lattice; the polarization is rational and need not be integral. Write m for the largest index with $F^m V_{\mathbf{O}} \neq 0$, and assume throughout that $\text{rk } F^m V_{\mathbf{O}} = 1$. We call this piece the *Hodge line*. On a simply connected flat chart its inclusion in $V_{\mathbf{O}}$ defines a holomorphic map

$$\ell : U_{\text{loc}} \longrightarrow \mathbb{P}(V_{\mathbf{C},u}).$$

Changes of flat chart compose ℓ with a constant projective linear transformation, so its differential rank is intrinsic.

2.1. The smallest rational subvariation. We call a lifted point *Hodge-generic* if it lies outside every proper Hodge locus of a rational flat tensor in tensor constructions on the variation and its dual. These are countably many closed analytic loci on the universal cover. Their complement is dense by the Baire theorem. Only this analytic genericity is needed in the Hodge arguments.

Lemma 2.1 (Rationally minimal variation). *There is a unique smallest rational subvariation $V^{\min} \subset V_{\mathbf{Q}}$ whose Hodge filtration contains $F^m V_{\mathbf{O}}$. At a Hodge-generic point, its fibre is the smallest rational Hodge substructure containing the Hodge line, and that fibre is simple. The subvariation has an induced polarization and integral lattice. On a smooth compactification with unipotent boundary monodromy, its extended Hodge line agrees with that of V .*

Proof. In a polarizable rational Hodge structure, intersect all rational Hodge substructures containing the specified line. A finite intersection already attains the minimum dimension, and gives the unique smallest one, say W . Semisimplicity shows that W is simple: in a decomposition into simple summands exactly one summand can have a nonzero deepest Hodge piece, since that piece has dimension one, and that summand already contains the line.

Here is also the generic-fibre justification, which is useful when passing between variations and individual Hodge structures. Trivialize the rational local system on the connected universal cover of U . For each rational endomorphism e , the condition that e preserve every F^p is closed analytic. Choose a lifted point outside all such loci which are proper. A rational Hodge projector onto W at this point therefore preserves the filtration everywhere on the cover. Its action on the Hodge line is the identity everywhere: it is an idempotent on a line bundle, and equals the identity at the chosen point. Its constant image is thus a subvariation containing the line. At every other point chosen in the same way, applying the argument in both directions shows that its image is again the smallest Hodge substructure containing the line. Uniqueness and the deck action now imply monodromy invariance, so this subvariation descends to U . Any rational subvariation containing the line contains it, by inspection at the chosen point.

Restriction of Q to a Hodge substructure is nondegenerate, and $V^{\min} \cap V_{\mathbf{Z}}$ supplies the lattice. The polarization gives a complementary subvariation whose deepest Hodge piece is zero. Canonical extensions preserve this direct sum, proving the last claim. This is the pure case of the transcendental-part construction of [5, Definition 2.9]. $\square$

2.2. From a fixed highest line to a fixed full period. A variation is *minimal for its highest line* if the generic fibre is the smallest rational Hodge substructure containing that line. Lemma 2.1 supplies this replacement without changing the line. The next two lemmas explain what this minimality buys. We use the usual Mumford–Tate group of a rational Hodge structure: the smallest rational algebraic group through which its Hodge homomorphism factors.

Lemma 2.2 (Normal subgroups and the deepest line). *Let $\mathbb{V}$ be a rationally polarized integral pure variation on a smooth connected complex algebraic variety, minimal for its deepest Hodge line. In a flat trivialization on the universal cover, let $G \subset \mathrm{GL}(V)$ be connected algebraic monodromy and let $P = \mathrm{MT}(V)$ be the generic Mumford–Tate group. If $H \subset G$ is a connected normal algebraic subgroup defined over $\mathbf{Q}$, and $V_{\mathbf{C}}^H$ contains the deepest line at a Hodge-generic lifted point, then H is trivial.*

Proof. The monodromy and fixed-part theorems give that G is semisimple and normal in the connected reductive group P ; see [3, §5, Theorem 1 and Corollary 1]. Over an algebraic closure, a connected normal subgroup of a semisimple group is the product of a collection of its almost simple factors. Conjugation by P preserves G . Since P is connected, it acts trivially on the finite set of these factors, and therefore preserves H .

Consequently V^H is a rational P -subrepresentation. It is a rational Hodge substructure at a Hodge-generic point, and it contains the deepest Hodge line. Minimality gives $V^H = V$. The inclusion $G \subset \mathrm{GL}(V)$ is faithful, so $H = 1$. $\square$

Lemma 2.3 (A flat automorphism over a fixed point). *Let U be a smooth connected complex algebraic variety with a rationally polarized integral pure variation $\mathbb{V}$ minimal for its deepest Hodge line. Suppose that an automorphism b of U fixes a point e , and that there is a flat isomorphism of variations*

$$\Phi_y : V_y \longrightarrow V_{b(y)}.$$

Use the fibre V_e as the flat reference space, and put $a = \Phi_e$. If, on a nonempty open subset of the universal cover, all deepest Hodge lines lie in a single eigenspace of a , then a preserves the complete Hodge filtration at every lifted point. In particular, locally at e the full lifted period maps at y and $b(y)$ are equal.

Proof. Write ρ for monodromy based at e . Flatness and $b(e) = e$ give

$$a\rho(\gamma)a^{-1} = \rho(b_*\gamma) \quad (\gamma \in \pi_1(U, e)).$$

Thus a normalizes both algebraic monodromy and its identity component G . Let λ be the eigenvalue in the hypothesis. The condition that the deepest line lies in $\ker(a - \lambda)$ is a holomorphic linear condition on the universal cover, so it holds throughout that connected cover by analytic continuation.

Fix a Hodge-generic lifted point and let $W \subset V_{\mathbf{C}}$ be the complex span of the monodromy orbit of its deepest line. This space is monodromy invariant, hence G -invariant. Equivariance places every line in this orbit among the deepest lines on the universal cover, so $a|_W = \lambda \mathrm{id}_W$. Every element $aga^{-1}g^{-1}$, with $g \in G$, therefore acts trivially on W , as do all its G -conjugates. Let H be their normal algebraic closure in G . It is defined over $\mathbf{Q}$, since a and G are rational. It is connected: the commutator map from the connected group G has connected image containing the identity, and the algebraic subgroup generated by its conjugates is the closure of products of

such connected sets. Although W need not be rational, V^H is rational and contains the chosen deepest line. Lemma 2.2 gives $H = 1$. Hence a centralizes G .

Pass to the connected finite étale cover $U_1 \rightarrow U$ corresponding to the inverse image of G under monodromy, and choose e_1 over e . Since a centralizes all monodromy on U_1 , it extends to a global flat rational section of $\text{End}(\mathbb{V}|_{U_1})$. The latter is a polarizable pure variation of weight zero. The flat section a has Hodge type $(0, 0)$ at e_1 , since Φ_e is a Hodge isomorphism. The base U_1 is a smooth algebraic variety and hence a Zariski open subset of a compact complex variety, so the fixed-part theorem applies. By [25, Theorem 7.22 and Corollary 7.23], a has type $(0, 0)$ everywhere. Thus a preserves the full filtration on the universal cover.

For completeness, take the lift of b to the universal cover fixing a chosen lift of e . Flatness of Φ gives $F^\bullet(b\tilde{y}) = aF^\bullet(\tilde{y})$ in the reference space. The assertion just proved makes the right-hand side equal to $F^\bullet(\tilde{y})$. $\square$

3. NUMERICAL RANK AND TANGENTIAL LIMITS

We now compare a boundary restriction of the extended highest line with the limiting projective line map. The Hodge-theoretic input identifies the limiting graded line. The analytic rank principle then decides what would happen if its tangential rank did not drop. For a nef line J on a smooth projective t -fold we use

$$\nu(J) = \max\{a \in \{0, \dots, t\} : c_1(J)^a H^{t-a} > 0\},$$

where H is ample. A line on a point has numerical dimension zero.

Let U and its variation be as in Section 2. Take a smooth projective compactification T of U with reduced simple normal crossings boundary D , and suppose that all local monodromies along D are unipotent. Denote the Deligne extension by $\overline{V}$ and the Schmid extension of the Hodge line by $J = F^m \overline{V}$. We use the following standard inputs.

Theorem 3.1 (Extension and positivity inputs). *Under these hypotheses:*

- (1) *The extended Hodge filtration is by subbundles of $\overline{V}$. For a morphism $g : (T', D') \rightarrow (T, D)$ of smooth projective varieties with reduced simple normal crossings boundaries and $g(T' \setminus D') \subset U$, the extended Hodge line of g^*V is g^*J . These extensions commute with direct sums and tensor products. If the monodromy logarithms of some boundary components vanish, the variation extends purely across those components, including their crossings away from the remaining boundary.*
- (2) *The line J is nef. It is big precisely when the top Kodaira–Spencer map*

$$T_U \longrightarrow \text{Hom}(F^m V_{\mathcal{O}}, F^{m-1} V_{\mathcal{O}} / F^m V_{\mathcal{O}})$$

is generically injective, equivalently when ℓ is generically immersive.

The first assertions are the unipotent extension and nilpotent-orbit theorems of [25, 11]; functoriality is also recorded in [5, Lemmas 2.16 and 5.6]. The second assertion is [4, Lemma 6.17].

The equivalence in the second assertion uses Griffiths transversality: the differential of the line map is the indicated Kodaira–Spencer map followed by the inclusion of F^{m-1}/F^m in $V_{\mathcal{O}}/F^m$.

Proposition 3.2 (Numerical rank). *If $r = \text{rk}_{\text{gen}} d\ell$, then*

$$(3.1) \quad \nu(J) = r.$$

Proof. Put $t = \dim T$, and fix an ample divisor H on T . For $1 \leq a \leq t$, choose a sufficiently general smooth complete intersection $C_a \subset T$ of dimension a , using a sufficiently large multiple of H . It meets the boundary transversely. At general points of $C_a \cap U$, the restriction of $d\ell$ to T_{C_a} has rank $\min(a, r)$. To see the required genericity, at a point where $d\ell$ has rank r , the a -planes with this property form a nonempty open subset of the tangent Grassmannian. A sufficiently

ample linear system prescribes the required first-order tangent conditions; Bertini then gives the assertion for general complete intersections. By Theorem 3.1, $J|_{C_a}$ is big exactly when $a \leq r$. Since it is nef, this is equivalent to $(J|_{C_a})^a > 0$, and hence to $c_1(J)^a H^{t-a} > 0$. This proves the formula, also when $r = 0$; dimension zero is immediate. $\square$

3.1. The limiting line along a divisor. We recall the boundary input in the form needed here. Let E be a component of D , let $E^\circ = E \setminus \text{Supp}(D - E)$, and let N be its monodromy logarithm. The monodromy weight filtration $W(N)$ is centred at w . The limiting filtration induces polarized pure variations on its graded pieces over E° , with unipotent monodromy at the remaining boundary. If k is the unique weight carrying the deepest line, then $J|_E$ is the Schmid extension of the top line of $\text{Gr}_k^{W(N)}$. These statements include compatibility at deeper strata; see [5, §2.5.2–§2.5.3 and Lemma 2.16(3)]. For a single N , the relevant primitive summand is polarized by the form induced by $Q(-, N^{k-w} -)$ when $k \geq w$.

The local expression below is Schmid's nilpotent-orbit normal form [25, Theorem 4.12]; its compatibility at crossings uses [11]. The rank comparison is the consequence needed here.

Proposition 3.3 (Boundary line). *On a transverse coordinate chart (u, v) about a general point of E , with $E = (u = 0)$, write a generator of the lifted Hodge line as*

$$(3.2) \quad e^{zN} a(u, v), \quad z = \frac{\log u}{2\pi i},$$

where a is holomorphic across $u = 0$ and $a(0, v) \neq 0$. After shrinking the chart there is an integer $\ell \geq 0$ such that

$$(3.3) \quad N^{\ell+1} a(0, v) = 0, \quad N^\ell a(0, v) \neq 0,$$

and the line belongs to weight $w + \ell$ in the limiting mixed Hodge structure. Moreover,

$$(3.4) \quad \nu(J|_E) \leq \text{rk}_{\text{gen}} d(v \mapsto [N^\ell a(0, v)]).$$

The map on the right is formed in the chosen flat coordinates. If $N = 0$, take $\ell = 0$; the variation then extends purely near the general point of E .

Proof. Put $a_0 = a(0, v)$. In the Deligne bigrading of the limiting mixed Hodge structure, F^m is a single one-dimensional summand $I^{m,q}$: all first indices are at most m , and $\dim F^m = 1$. Set $\ell = m + q - w$. The primitive decomposition on weight $w + \ell$ expresses each summand as a sum of terms

$$N^j P_{w+\ell+2j}, \quad j \geq \max(0, -\ell),$$

where P_{w+b} denotes the primitive part for $b \geq 0$. A contribution to Hodge type (m, q) with $j > 0$ would come from type $(m + j, q + j)$, impossible because its first index exceeds m . Thus $j = 0$, $\ell \geq 0$, and the top line is primitive. Hard Lefschetz for the monodromy weight filtration gives nonvanishing of its image under N^ℓ and vanishing under $N^{\ell+1}$ on the associated graded. Since N has bidegree $(-1, -1)$ in the Deligne bigrading, these statements hold for a_0 itself: a vector in a single Deligne summand maps injectively to its weight-graded piece. The integer ℓ is constant on a sufficiently small open stratum, so the identities hold throughout the chosen v -chart.

Let J_E^{gr} be the extended top line of the pure variation on $\text{Gr}_{w+\ell}^{W(N)}$ along E° . The boundary extension input identifies it with $J|_E$. Projecting $N^\ell a_0$ to $\text{Gr}_{w-\ell}^{W(N)}$ yields the image of this graded top line under the flat isomorphism

$$N^\ell : \text{Gr}_{w+\ell}^{W(N)} \longrightarrow \text{Gr}_{w-\ell}^{W(N)}.$$

Consequently the projective map of the graded line has rank no greater than $v \mapsto [N^\ell a_0(v)]$: projection is a fixed linear map on the flat chart, defined near the line in question. Apply

Proposition 3.2 to the graded variation on E° and its smooth projective compactification E . This proves (3.4). $\square$

3.2. A local rank principle. The boundary-line proposition has reduced the numerical question to a projective differential. We next prove the local assertion that will force a symmetry to fix the interior whenever the limiting rank is as large as the interior rank. This analytic fact requires control of tangential derivatives only. A sector below has the form $S_\epsilon = \{0 < |u| < \epsilon, \alpha < \arg u < \beta\}$ with a fixed branch of the argument and finite opening $\beta - \alpha < 2\pi$.

Lemma 3.4 (Rank and a tangential limit). *Let $B \subset \mathbf{C}^n$ be a polydisc and let $f : S_\epsilon \times B \rightarrow M$ be a holomorphic map to a complex manifold. Suppose $f(u, \cdot)$ converges uniformly on compact subsets of B , as $u \rightarrow 0$ throughout the sector, to a holomorphic map f_0 . Assume that the full generic differential rank of f is at most r and that $\text{rk } df_0(v_0) = r$ at some $v_0 \in B$. Then, after shrinking the sector and taking neighborhoods $v_0 \in B' \Subset B_0 \subset B$,*

$$f(S_{\epsilon'} \times B') \subset f_0(B_0),$$

where the indicated local image of f_0 is an embedded complex submanifold of dimension r . In particular, every fixed closed analytic subset containing this local image also contains the nearby image of f .

Proof. Use a common coordinate chart in M , shrinking B and the sector. Cauchy estimates give convergence of all derivatives in the B variables on smaller polydiscs. The $(r+1)$ -minors of df vanish identically, so those of df_0 vanish as well. Thus f_0 has constant rank r near v_0 . The holomorphic constant rank theorem supplies source coordinates $v = (x, y) \in \mathbf{C}^r \times \mathbf{C}^{n-r}$ and target coordinates in which $f_0(x, y) = (x, 0)$.

Write $f = (F, G)$ in these coordinates, with $F(u, x, y) = x + E(u, x, y)$. On fixed smaller polydiscs, both E and $\partial_x E$ tend uniformly to zero. Choose nested x -balls, a smaller ball W for w , and a ball Y for y . For sufficiently small ϵ' , the maps

$$x \mapsto w - E(u, x, y)$$

send a fixed closed x -ball into itself and have Lipschitz constant less than $1/2$, uniformly for $(u, w, y) \in S_{\epsilon'} \times W \times Y$. The contraction theorem gives a unique solution $x = \chi(u, w, y)$ of $F(u, x, y) = w$ on this common product. The holomorphic inverse function theorem and uniqueness make χ holomorphic in all interior parameters; moreover $\chi \rightarrow w$ uniformly on smaller products.

In the new source coordinates the map is

$$f(u, \chi(u, w, y), y) = (w, H(u, w, y)).$$

The w -derivatives already have rank r . The rank bound therefore forces $\partial_u H = 0$ and $\partial_y H = 0$: either derivative, if nonzero, would give an additional image vector with zero first r coordinates. Since the sector and Y are connected, H depends only on w . Its limit as $u \rightarrow 0$ is zero, hence $H = 0$. Shrinking the original source neighborhood so that its F -values belong to W proves the assertion. If $r = 0$, the rank bound directly makes f constant on the connected source, and the same conclusion follows. $\square$

Lemma 3.5 (Nilpotent tangential convergence). *Let N be a nilpotent endomorphism of a finite-dimensional complex vector space, and let $a(u, v)$ be holomorphic on $\Delta \times B$. Suppose, for a fixed $\ell \geq 0$, that $N^{\ell+1}a(0, v) = 0$ and $N^\ell a(0, v) \neq 0$ throughout B . On a sector with the above choice of $z = \log u / (2\pi i)$,*

$$(3.5) \quad [e^{zN} a(u, v)] \longrightarrow [N^\ell a(0, v)]$$

normally on sufficiently small v -charts, with all tangential derivatives. Thus Lemma 3.4 applies whenever the rank of this limit equals an upper bound for the full rank of the line map. Its nearby image then lies in $\mathbb{P}(\ker N)$.

Proof. Choose k with $N^{k+1} = 0$, and write $a(u, v) = a_0(v) + uq(u, v)$. Set $b(v) = N^\ell a_0(v)/\ell!$. Direct expansion gives

$$z^{-\ell} e^{zN} a(u, v) = b(v) + \sum_{j < \ell} \frac{z^{j-\ell}}{j!} N^j a_0(v) + u \sum_{j=0}^k \frac{z^{j-\ell}}{j!} N^j q(u, v).$$

For every fixed tangential derivative, uniformly on compact subsets of B , the remainder is

$$O(|z|^{-1}) + O(|u| |z|^{k-\ell});$$

the first term is absent when $\ell = 0$. On the sector $|z|$ is comparable to $|\log |u||$, so both terms tend to zero. Choose a linear functional nonzero on $b(v_0)$ and shrink the v -chart. Dividing by that functional gives convergence with all tangential derivatives in a fixed affine projective chart, proving (3.5). Finally $Nb = 0$, so the limiting image belongs to the fixed projective linear subspace $\mathbb{P}(\ker N)$, and the last assertion follows from Lemma 3.4. $\square$

Remark 3.6. All numerical-rank statements are unchanged on replacing a Hodge line by a positive rational multiple of its class in $\text{Pic}(T) \otimes \mathbf{Q}$. The local statements also apply with $N = 0$, when the line map extends holomorphically across the divisor.

4. RANK LOSS ON THE BOUNDARY AND THE RAMIFICATION DIVISOR

The preceding lemmas apply to a smooth model of a full period image. At a neat level the variation is unipotent at the boundary. Returning to the original level also introduces ramification divisors. We now prove the strict numerical rank drop for both kinds of divisor.

We specify the finite-group data because the group acting on the variation can be larger than the effective group acting on its base. Let T be a smooth connected projective complex variety of positive dimension, let D_T be a reduced simple-normal-crossing divisor, and put $U = T \setminus D_T$. Let $\mathbb{V}$ be a rationally polarized integral pure variation on U , minimal for its entire highest Hodge line, with unipotent boundary monodromy. Write J for the extended highest line. Let D_N be the union of the components of D_T with nonzero monodromy logarithm, and put $U_N = T \setminus D_N$. The variation extends purely over U_N by Theorem 3.1.

Suppose a finite group H acts on T , preserves D_T , and acts on the variation by flat Hodge isomorphisms preserving its integral lattice and rational polarization and covering that action. Set

$$G_f = H / \ker(H \longrightarrow \text{Aut}(T)).$$

Thus G_f acts effectively on T . We retain H for its action on the variation, since its kernel on T can act nontrivially on fibres. Uniqueness of the pure extension preserves this action over U_N .

Fix a connected period domain $\mathcal{D}$ for the rational polarized reference space and a neat arithmetic group Γ' of lattice isometries containing the monodromy. Let

$$\varphi : U_N \longrightarrow \Gamma' \backslash \mathcal{D}$$

be the full period map. Assume that the normalization B of the closure of its actual image is an algebraic variety, that the natural rational map $T \dashrightarrow B$ is birational, and that φ is generically immersive. In particular,

$$\mathbf{C}(T) = \mathbf{C}(B).$$

These assumptions describe the actual period-image model. The algebraicity of B is the period-image theorem of [4, Theorem 1.1]; here its construction is an explicit input to the rank-loss statement.

Theorem 4.1 (Boundary and ramification rank loss). *For the data just specified, put $r = \nu(J)$. Then $r > 0$. If E is either a component of D_N , or a component of D_T fixed pointwise by a nonidentity element of G_f , then*

$$\nu(J|_E) < r.$$

In particular, after placing the divisorial fixed loci in D_T on an equivariant smooth model, the conclusion applies to every divisorial ramification component of $T \rightarrow T/G_f$.

Proof. Proposition 3.2 identifies r with the generic rank of the highest-line map. We first show that this rank is positive, then consider the nilpotent and finite cases.

If $r = 0$, the highest line is constant on the connected universal cover of U_N . Monodromy preserves it. Its connected algebraic monodromy group G is semisimple and hence acts trivially on that line. Lemma 2.2, applied with the normal subgroup equal to G , gives $G = 1$. After a finite étale cover the monodromy is trivial, and the fixed-part theorem makes the entire variation constant. Its full period map then has rank zero, contrary to generic immersivity on the positive-dimensional variety T . Thus $r > 0$.

A component with nonzero monodromy logarithm. Let $E \subset D_N$ and write $N \neq 0$ for its rational monodromy logarithm. In coordinates (u, v) near a general point of E , with $E = (u = 0)$, the lifted highest-line map has the form

$$[e^{zN} a(u, v)], \quad z = \frac{\log u}{2\pi i}.$$

Proposition 3.3 provides an integer $\ell \geq 0$ with

$$N^{\ell+1} a(0, v) = 0, \quad N^\ell a(0, v) \neq 0, \quad \nu(J|_E) \leq \text{rk}_{\text{gen}} d(v \mapsto [N^\ell a(0, v)]).$$

Lemma 3.5 gives its normal tangential convergence on a sector from the interior line maps. All interior $(r + 1)$ -minors vanish. Cauchy convergence of tangential derivatives therefore bounds the rank of the limiting map by r .

Suppose that $\nu(J|_E) \geq r$. The limiting map has rank exactly r at a suitable general point, and its image lies in $\mathbb{P}(\ker N)$. Lemma 3.4 forces all nearby interior lines to lie in this same linear subspace. The condition that N annihilates the highest line is holomorphic, so it continues over the connected universal cover of U_N .

The one-parameter group $\exp(tN)$ belongs to G : it is the connected Zariski closure of the cyclic unipotent local monodromy. Every monodromy conjugate of this group fixes the highest line pointwise. Let G_N be the algebraic subgroup they generate. It is connected, defined over $\mathbf{Q}$, and normalized by monodromy; Zariski density makes it normal in G . Its invariant space contains the highest line at a Hodge-generic point. By Lemma 2.2, $G_N = 1$, contradicting $N \neq 0$. This proves the required inequality for components of D_N .

We have controlled all divisors at which the pure variation fails to extend. The remaining obstruction is a divisor of finite ramification where the variation does extend. Its inertia element can act on the rational fibre even when it fixes every point of the divisor, so we must use the full-period conclusion of Lemma 2.3.

A divisor fixed by finite inertia. Let $b \in G_f$ be nonidentity and fix E pointwise. If E is contained in D_N , the preceding case applies. Otherwise a general point $e \in E$ lies in U_N . Choose a lift $h \in H$ of b . The retained H -action supplies a flat Hodge isomorphism

$$\Phi_y : V_y \longrightarrow V_{b(y)}$$

on U_N . Use V_e as a flat reference space on a small simply connected b -invariant neighborhood, and set $a = \Phi_e$. If x is the full lifted period map, flatness gives

$$x(by) = a x(y).$$

The operator a is a rational Hodge isometry at e . Because H acts on the variation and $b(e) = e$, the group-action law gives $a^{\text{ord}(h)} = 1$. Since b fixes E , every highest line on the nearby part of E is an eigenline of a . Its eigenvalue is constant on a connected small patch, so these lines lie in one eigenspace $\ker(a - \lambda)$.

Proposition 3.3 with $N = 0$ bounds $\nu(J|_E)$ by the rank of this extending line map on E .

Suppose again that $\nu(J|_E) \geq r$. The rank on E is at most r , since all $(r + 1)$ -minors of the ordinary extending line map vanish by continuation from the interior. It is therefore r . Apply Lemma 3.4 in a transverse coordinate with $N = 0$. A nonempty open set of lifted highest lines lies in $\ker(a - \lambda)$. Lemma 2.3, applied on the smooth algebraic variety U_N with fixed point e , now shows that a preserves every full period. Consequently $x(by) = x(y)$.

Projecting to $\Gamma' \backslash \mathcal{D}$ gives $\varphi(by) = \varphi(y)$ on a nonempty open set. Since B is the normalization of the actual image, its normalization map is birational and hence injective on a dense open subset of that image. It follows that b acts trivially on $\mathbf{C}(B)$. The chosen model has $\mathbf{C}(T) = \mathbf{C}(B)$, so b acts trivially on T , contrary to its nonidentity in G_f . This proves the strict inequality.

Finally, in a finite quotient in characteristic zero, ramification at a divisorial valuation is detected by a nontrivial inertia group, which acts identically on that divisor's function field. Its elements therefore fix the divisor pointwise. The last assertion follows from the case just proved. $\square$

Remark 4.2. The conclusion concerns the numerical dimension of the highest line, while generic immersivity is a hypothesis on the full period map. Neither the proof nor the conclusion identifies these ranks. The function-field equality with the actual period image is used only at the last contradiction in the finite case; it cannot be replaced there by an arbitrary finite extension of that function field.

5. THE PERIOD QUOTIENT AND ADJOINT BIGNESS

We prove Theorem 1.2. The full period image supplies an algebraic base, and Section 4 controls the divisors introduced by compactification and finite descent. We first show that those numerical rank drops suffice to remove the divisors from an adjoint class. This is a section-growth calculation; no abundance statement for the highest line is needed.

5.1. Section estimates.

Lemma 5.1. *Let T be a smooth projective t -fold, A a rational divisor, and J a nef rational divisor with $r = \nu(J) > 0$, and $D = \sum_i d_i E_i$ an effective integral divisor with smooth prime components. Suppose*

$$\nu(J|_{E_i}) < r \quad \text{for every } i.$$

If $A + D$ is big, then $A + jJ$ is big for every sufficiently large integer j .

Proof. Choose an ample rational divisor H_1 such that $A + D - H_1$ is effective. In this proof m tends to infinity through sufficiently divisible positive integers. For each fixed $j \geq 0$, asymptotic Riemann–Roch for the ample divisor $H_1 + jJ$ gives

$$\liminf_{m \rightarrow \infty} \frac{t!}{m^t} h^0(T, m(A + D + jJ)) \geq (H_1 + jJ)^t \geq cj^r$$

for a constant $c > 0$ and all $j \geq 1$. The last inequality follows from $J^r H_1^{t-r} > 0$.

Remove mD one prime divisor at a time. Each quotient in a divisor exact sequence is a line bundle on some E_i , of the form

$$\mathcal{O}_{E_i} \left(m(A + D + jJ) - \sum_h k_h E_h \right), \quad 0 \leq k_h \leq md_h.$$

There is a single ample rational divisor H_2 on T for which every such quotient has at most $h^0(E_i, m(H_2 + jJ)|_{E_i})$ sections, in divisible degrees and for every $j \geq 0$. To see this, choose an ample Cartier divisor B_0 with a globally generated multiple clearing the denominator of $B_0 - (A + D)$, and ample Cartier divisors B_h such that both B_h and $B_h + E_h$ are globally generated. Enlarge B_0 if necessary and take $H_2 = B_0 + \sum_h d_h B_h$. The difference, restricted to E_i , is

$$m(B_0 - A - D) + \sum_h ((md_h - k_h)B_h + k_h(B_h + E_h)).$$

It has a section nonzero on E_i , by global generation and multiplication, giving the claimed bound.

There are $m \sum_i d_i$ quotient terms. For fixed j , asymptotic Riemann–Roch on each E_i bounds their total contribution on the scale $m^t/t!$ by

$$t \sum_i d_i (H_2 + jJ)^{t-1} \cdot E_i.$$

This is $O(j^{r-1})$, since every intersection containing at least r factors of $J|_{E_i}$ vanishes. Consequently

$$\liminf_{m \rightarrow \infty} \frac{t!}{m^t} h^0(T, m(A + jJ)) \geq cj^r - O(j^{r-1}) > 0$$

for all sufficiently large j , proving bigness. $\square$

Lemma 5.2. *Let $\pi : T \rightarrow S$ be a surjective generically finite morphism of smooth projective varieties. Let $E \geq 0$ be a rational divisor on T whose components have image of codimension at least two in S . For a rational divisor D on S , the divisor $\pi^*D + E$ is big if and only if D is big.*

Proof. Factor π as $T \xrightarrow{\rho} \bar{S} \xrightarrow{\tau} S$, where $\bar{S}$ is the normalization of S in $\mathbf{C}(T)$. Then ρ is birational and τ is finite. Every component of E is ρ -exceptional, since a finite map preserves dimensions. Normality gives $\rho_* \mathcal{O}_T(mE) = \mathcal{O}_{\bar{S}}$ in divisible degrees: a rational function with poles only on exceptional divisors has no codimension-one pole downstairs. Projection formula identifies the sections of $m(\pi^*D + E)$ with those of $m\tau^*D$. Finally, bigness is invariant under surjective finite pullback. $\square$

5.2. Construction and descent.

Proof of Theorem 1.2. We may first make the birational modification allowed in the hypothesis. Replace $\mathbb{V}$ by the minimal rational subvariation from Lemma 2.1, and equip it with the intersection lattice and the restricted polarization. Its deepest Hodge line and the line $J_{\widehat{\mathbb{V}}}$ are unchanged. We continue to denote this variation by $\mathbb{V}$.

The actual period image and its finite quotient. Choose a rational reference space with lattice and polarization, and let $\mathcal{D}$ be the connected period-domain component containing a chosen lift of the periods. Let Γ be the arithmetic group of integral polarizing isometries preserving this component; it contains the monodromy of $\mathbb{V}$. Let $\Gamma' \triangleleft \Gamma$ be a normal neat subgroup of finite index. The kernel of

$$\pi_1(U) \longrightarrow \Gamma/\Gamma'$$

defines a connected finite étale Galois cover $U' \rightarrow U$. Denote its deck group by H ; thus H is the image of this homomorphism, not necessarily all of Γ/Γ' .

The period map of $\mathbb{V}|_{U'}$ factors algebraically through the normalization B of the closure of its image in $\Gamma' \backslash \mathcal{D}$. The space B is a normal quasiprojective variety, and on a smooth dense open subset its period realization has generically immersive differential; this is the period-image theorem [4, Theorem 1.1]. Here and below the function field of B is the function field of this normalized *actual image*. We do not replace it by its relative algebraic closure in $\mathbf{C}(U')$.

Neatness makes the arithmetic action free. The universal local system on the quotient is

$$(\mathcal{D} \times V_{\mathbf{Z}})/\Gamma',$$

with its tautological filtration. On the smooth open of B this filtration is horizontal: the dominant map $U' \rightarrow B$ is generically submersive and its pullback is the original variation, so Griffiths transversality holds on a dense open and hence everywhere. Thus the restricted local system and filtration form a polarizable integral variation. Moreover, the action of Γ/Γ' on this local system is an honest action:

$$[x, v] \mapsto [\gamma x, \gamma v].$$

Normality of Γ' makes the formula independent of the representative γ . Restricting gives an action of H on the variation over B , covering its action on B . The action on B is algebraic: apply uniqueness in [4, Theorem 1.1] to the period map and its expression obtained by a deck transformation on U' and the corresponding arithmetic translation on the target, and then lift to the normalization. Let

$$G_f = H / \ker(H \rightarrow \text{Aut}(B))$$

be the effective group on the image. The full group H will still be retained for its action on the variation.

Choose a G_f -equivariant smooth projective compactification S' of a smooth dense invariant open of B , with snc boundary, by equivariant compactification and resolution. All boundary monodromies of its variation are unipotent. Indeed they are quasi-unipotent by the monodromy theorem, and they belong to the neat group Γ' ; their root-of-unity eigenvalues must therefore all be one. Let J' denote the extended deepest Hodge line and put $L' = cJ'$. This is nef by Theorem 3.1. The H -action extends to J' by functoriality of the unipotent Deligne and Hodge extensions.

Let $q : S' \rightarrow Q_0 = S'/G_f$ be the finite quotient. Although an element of $\ker(H \rightarrow G_f)$ may act nontrivially on J' , it acts on its fibres through finite-order characters. Taking a tensor power divisible by $|H|$ kills the fibre action of every stabilizer in H , including this kernel. That tensor power therefore descends to an actual line bundle on Q_0 by finite-group descent. Consequently there is a rational line bundle $\bar{L}$ on Q_0 with

$$q^* \bar{L} \sim_{\mathbf{Q}} L'.$$

It is nef, since its pullback by the finite surjective map q is nef. Choose a smooth projective resolution $s : S_0 \rightarrow Q_0$ and put $L_0 = s^* \bar{L}$.

The equivariant period map on U' descends to a dominant rational map $Y \dashrightarrow Q_0$, and hence to S_0 . Resolve its graph by a smooth projective birational modification of Y , obtaining

$$p_0 : Y \rightarrow S_0.$$

We claim that

$$(5.1) \quad M \sim_{\mathbf{Q}} p_0^* L_0.$$

To verify this claim, choose a common smooth projective alteration W dominating the alteration in the hypothesis and the level cover, and resolve the graph of its rational map to S' . Write $\mu : W \rightarrow Y$ and $t : W \rightarrow S'$ for the resulting morphisms. Arrange snc boundaries by further resolution, containing the inverse images of the boundaries in question. The pulled-back variations agree on a dense open subset. Unipotent Deligne extensions and their extended Hodge filtrations commute with these pullbacks between smooth snc pairs; see [5, Lemma 5.6], or the unipotent extension theory [25, 11]. The extended top line J_W is consequently the pullback of the top line on either model. The hypothesis and the commutative maps to Q_0 give

$$\mu^* M \sim_{\mathbf{Q}} cJ_W \sim_{\mathbf{Q}} t^* L' \sim_{\mathbf{Q}} \mu^* p_0^* L_0.$$

Pullback by a proper generically finite map is injective on rational line-bundle classes: applying divisor pushforward, or the norm of a principal rational function, multiplies the original class by the degree. This proves (5.1). In particular no boundary twist has been inferred from an equality merely on the open locus.

If $\dim S_0 = 0$, equation (5.1) gives $M \sim_{\mathbf{Q}} 0$, and the theorem follows. Assume henceforth that $\dim S_0 > 0$.

A common equivariant model and log general type. Normalize S_0 in $\mathbf{C}(S')$, and take a G_f -equivariant smooth projective resolution of the resulting model and its birational map to S' . Denote the resulting maps by

$$\eta : T \longrightarrow S', \quad \pi : T \longrightarrow S_0.$$

Thus η is birational, π is generically finite with Galois function-field group G_f , and G_f acts on T over S_0 . Choose the resolution so that there is a G_f -invariant snc boundary D_T containing the complement of the open period locus and the support of the Jacobian divisor of π . The pulled-back variation has unipotent boundary monodromy. If J_T is its extended top line, then

$$(5.2) \quad L_T := cJ_T \sim_{\mathbf{Q}} \eta^* L' \sim_{\mathbf{Q}} \pi^* L_0.$$

Let D_N be the reduced sum of those components of D_T having nonzero monodromy logarithm. The variation extends as a pure polarized variation over

$$U_N = T \setminus D_N.$$

Indeed, near a point where all the remaining logarithms are zero, the nilpotent-orbit extension is a pure polarized Hodge filtration; the several-variable extension gives the same statement at crossings [25, 11]. These extensions are unique and preserve the H -equivariance of the variation. The divisor D_N is snc, since it is a union of components of D_T .

The full period map on U_N has generically immersive differential, because T is birational to B . The log-general-type theorem for a polarized variation with generically immersive period map therefore gives

$$(5.3) \quad K_T + D_N \quad \text{big}.$$

Here the applicable theorem is [7, Theorem 1.1]. Its boundary is precisely D_N , because the pure variation has just been extended over all the other boundary components.

We now have precisely the geometric data of Theorem 4.1: T is birational to the actual normalized period image, the full period map is generically immersive, and the retained finite group H acts on its integral polarized variation. The effective group on T is G_f . Minimality is preserved here: the variation on T and the original variation have the same Hodge structures at Hodge-generic points, since the original source dominates the period image. Consequently

$$(5.4) \quad r := \nu(L_T) > 0, \quad \nu(L_T|_E) < r$$

for every component of D_N and every divisorial ramification component of π mapping onto a divisor of S_0 . For the latter, the inertia group of the Galois extension $\mathbf{C}(T)/\mathbf{C}(S_0)$ contains a nonidentity element of G_f fixing the divisor pointwise. These fixed loci were included in D_T . Positive rational rescaling from J_T to L_T leaves all these numerical dimensions unchanged. The use of the actual image, before Stein factorization, is exactly the function-field hypothesis needed in the finite-inertia part of Theorem 4.1.

Removal of the boundary and ramification. Write the effective Jacobian divisor as

$$R_\pi = K_T - \pi^* K_{S_0} = R_{\text{div}} + R_{\text{exc}},$$

where R_{div} is supported on components mapping onto divisors of S_0 , and each component of R_{exc} has image of codimension at least two. In the volume Lemma 5.1, take

$$J = L_T, \quad D = D_N + R_{\text{div}}, \quad A = K_T - R_{\text{div}} = \pi^* K_{S_0} + R_{\text{exc}}.$$

All components of D satisfy the rank inequality above, and $A + D = K_T + D_N$ is big by (5.3). The volume lemma gives

$$A + jL_T \sim_{\mathbf{Q}} \pi^*(K_{S_0} + jL_0) + R_{\text{exc}} \quad \text{big for } j \gg 0.$$

Lemma 5.2 removes the effective divisors exceptional over S_0 and descends bigness through the generically finite map. Therefore

$$(5.5) \quad K_{S_0} + jL_0 \quad \text{is big for every sufficiently large integer } j.$$

Connected fibres. Finally take the Stein factorization of p_0 , and resolve its finite normal intermediate variety to a smooth projective variety S . Resolving the induced rational map from Y gives, by further smooth projective birational modification, a morphism $p : Y \rightarrow S$ with connected fibres. Indeed, its Stein factor is finite birational over the normal variety S , because $\mathbf{C}(S)$ is already algebraically closed in $\mathbf{C}(Y)$, and is therefore S itself. If $\sigma : S \rightarrow S_0$ is the resulting generically finite morphism, set $L = \sigma^* L_0$. It is nef, and (5.1) pulls back to $M \sim_{\mathbf{Q}} p^* L$. Since both S and S_0 are smooth, the Jacobian divisor of σ is effective. Thus

$$K_S + jL = \sigma^*(K_{S_0} + jL_0) + R_\sigma$$

is big for $j \gg 0$ by (5.5) and preservation of bigness under generically finite pullback. This proves the theorem. $\square$

6. CANONICAL BUNDLE DATA AND THE LEAST-INDEX ROOT COVER

We return to the original algebraic fibre space. The goal of this section is to construct the rational line to which Theorem 1.2 applies and to retain the exact section comparison with the source. The generic fibre of a relative Iitaka fibration has Kodaira dimension zero; its least-index canonical root cover provides the entire highest Hodge line.

We work over $\mathbf{C}$. All varieties in this section are projective. A rational pair (Y, B) is klt if, on a log resolution, the boundary defined by crepant pullback of $K_Y + B$ has every coefficient less than one; it is log canonical if those coefficients are at most one. We allow negative coefficients when explicitly discussing a sub-pair. A b-divisor records compatible divisor traces on all birational models of the base. Saying that its moduli part descends to Y means that every higher trace is the pullback of its trace M on Y . We distinguish a divisor exceptional over a birational model of the total space from a divisor whose image on the base has codimension at least two.

6.1. The canonical bundle formula and its normalization. We use the following parabolic form of the canonical bundle formula as a deep theorem.

Theorem 6.1 (Parabolic canonical bundle formula). *Let $g_0 : X_0 \rightarrow Y_0$ be an algebraic fibre space between smooth projective varieties, with geometric generic fibre of Kodaira dimension zero. After birational modifications of its source and base there is a commutative diagram*

$$\begin{array}{ccc} X' & \xrightarrow{\alpha} & X_0 \\ g \downarrow & & \downarrow g_0 \\ Y & \xrightarrow{\beta} & Y_0 \end{array}$$

with X' and Y smooth, and rational divisors B, M, R such that

$$(6.1) \quad K_{X'} \sim_{\mathbf{Q}} g^*(K_Y + B + M) + R.$$

Here $B \geq 0$, its support has simple normal crossings, (Y, B) is klt, and M is nef. Moreover,

$$(6.2) \quad g_* \mathcal{O}_{X'}(\lfloor kR^+ \rfloor) = \mathcal{O}_Y \quad (k \in \mathbf{Z}_{>0}), \quad \text{codim}_Y g(\text{Supp } R^-) \geq 2,$$

and R^- is α -exceptional. The divisor M is the trace of the moduli b -divisor, which can be assumed to descend to Y . On a sufficiently high generically finite base change admitting semistable reduction, its pullback is $\mathbf{Q}$ -linearly equivalent to a positive rational multiple of the Schmid extension of the distinguished canonical Hodge eigensheaf of the canonical root cover.

The residual divisor and discriminant assertions, with the normalization by the least nonzero pluricanonical index, are [15, §4.4 and Theorem 4.5(i)–(v)]. For original boundary zero, the logarithmic semistable part agrees with the semistable part of Section 2 by [15, Proposition 4.7]. The moduli interpretation follows from the discriminant-threshold definition in [15, Definition 4.3] and the following Ambro results. The b -divisor and Hodge assertions use [1, Theorems 0.2 and 4.4(1),(3), Lemma 5.2(4)–(5), and Propositions 5.4–5.5]; see also [2, Proposition 3.1]. These statements allow the auxiliary sub-boundary to have negative coefficients.

We record how the hypotheses of these results fit together. In the notation of [15, §4.4], applied with original boundary zero, the crepant boundary upstairs is the negative of the effective relative canonical divisor over the smooth original source. Thus the choice $\Xi = 0$ satisfies the hypotheses of their Theorem 4.5, including those of part (v). The coefficients of our B are their s_P/b ; part (v) places them in $[0, 1)$. Their residual divisor is bR . We verify the all-degree direct-image condition from the normalization below, so that it remains available after the final change of model. For the exceptionality assertion, flatten before resolving the main transform, as in [15, §4.4]. The inverse image of a base subset of codimension at least two has codimension at least two in the flat main transform; finite normalization preserves this bound. A prime divisor of its resolution lying over that subset is therefore exceptional over the normalized main transform. This transform maps birationally to X_0 , so the prime is also α -exceptional. Apply this to every component of R^- . Its small image on Y alone would not imply exceptionality over X_0 .

Here is also the relevant normalization of the associated sub-pair. On the geometric generic fibre C , let b be the least positive integer with $H^0(C, bK_C) \neq 0$, and let D_C be the divisor of its unique section up to scalar. The horizontal restriction is

$$R_C = \frac{1}{b} D_C, \quad \Delta_C = -R_C.$$

The least positive integer making $K_C + \Delta_C$ principal is also b . Indeed, $bK_C - D_C$ is principal. Conversely, if $m(K_C - D_C/b)$ is principal for a positive integer m , then mD_C/b is an integral effective divisor linearly equivalent to mK_C . Thus $H^0(C, mK_C) \neq 0$, which forces $m \geq b$. This identifies the least-index normalization in the sub-pair construction with the pluricanonical index used for the root cover. A rational relative pluricanonical section first defines a divisor R_0 with this restriction. For a prime divisor $P \subset Y$, subtract from R_0 the pullback of P multiplied by

$$a_P = \min_{E \mapsto P} \frac{\text{coeff}_E(R_0)}{\text{mult}_E(g^*P)}.$$

Only finitely many a_P are nonzero. The resulting residual divisor is nonnegative over the generic point of P , with coefficient zero on at least one component over P .

This normalization also gives the direct-image condition in every positive degree. Let $V \subset Y$ be a nonempty open subset, let $k > 0$ be an integer, and let φ represent a section of $\mathcal{O}_{X'}(\lfloor kR^+ \rfloor)$ over $g^{-1}(V)$. On the geometric generic fibre C , its poles are bounded by $\lfloor kD_C/b \rfloor$, hence by an integral multiple of D_C . Since $D_C \sim bK_C$ and $\kappa(C) = 0$, every such function is constant on C . Since $g_* \mathcal{O}_{X'} = \mathcal{O}_Y$, the field $\mathbf{C}(Y)$ is algebraically closed in $\mathbf{C}(X')$. Thus $\varphi = g^* \psi$ for a rational function ψ on V . For each prime divisor P meeting V , choose a component E over P on which

the normalized residual coefficient is zero. Regularity of φ along E gives

$$0 \leq \text{ord}_E(g^*\psi) = \text{mult}_E(g^*P) \text{ord}_P(\psi).$$

Thus ψ has no codimension-one pole on V , and normality of Y makes it regular. Pullbacks of regular functions give the reverse inclusion, proving

$$g_*\mathcal{O}_{X'}(\lfloor kR^+ \rfloor) = \mathcal{O}_Y \quad (k \in \mathbf{Z}_{>0}).$$

For the sub-pair $(X', -R)$ its discriminant coefficient is $1 - t_P$, where

$$t_P = \sup\{t \in \mathbf{R} : (X', -R + t g^*P) \text{ is sub-lc over the generic point of } P\}.$$

Nonnegativity of R there implies $t_P > 0$; the component with residual coefficient zero gives $t_P \leq 1$. This explains the effective boundary in this normalization.

The rank-one hypothesis for this sub-pair holds even though Δ_C is negative. Indeed, $\kappa(C, R_C) = 0$. On a log resolution of $(C, -R_C)$, a section of the rounded discrepancy sheaf pushes down to a rational function on C whose poles are bounded by a sufficiently large integral multiple of D_C . The space of such functions has dimension one, while the constant function is allowed. Consequently

$$\text{rk } g_*\mathcal{O}_{X'}(\lceil \mathbf{A}(X', -R) \rceil) = 1,$$

where the expression is interpreted by pushing down from a log resolution. This is the rank condition in [1, introductory assumptions (1)–(3)].

After choosing a model on which the moduli b-divisor descends, take a common higher base model satisfying the flattening and normal-crossing requirements, with boundary containing the total transform of the original normalized discriminant support and all exceptional base divisors. Crepantly pull back the auxiliary sub-pair to a smooth resolved main transform, taking a log resolution that includes its residual support and the pullbacks of the base boundary. Apply the same minimum-coefficient normalization to this residual divisor. Its horizontal restriction is again D_C/b for the unique least-index pluricanonical divisor on the new geometric generic fibre: for the induced birational morphism $\lambda : C' \rightarrow C$, crepant pullback gives

$$\lambda^*(D_C/b) + K_{C'/C} = D_{C'}/b.$$

The preceding argument therefore restores the all-degree direct-image condition on this final model. The normalized residual is nonnegative over every codimension-one base point; its negative part has base image of codimension at least two and is exceptional over the original source by the flattening argument above. Over the strict transform of an original base prime, its zero-coefficient component survives and the crepant residual stays nonnegative, so renormalization makes no further change there. Any new discriminant support is therefore exceptional and lies in the chosen simple-normal-crossing boundary. Its coefficients are in $[0, 1)$ by the same threshold calculation, so the discriminant is effective and klt.

This renormalization does not change the moduli b-divisor. Indeed, changing a sub-boundary from Δ to $\Delta + g^*A$ changes the divisor pulled back in $K_{X'} + \Delta$ by A , while the discriminant changes by A as well, by its threshold definition. Their difference is unchanged, and the same calculation holds on every higher base model. Crepant birational pullback likewise preserves the induced b-divisor. Thus the final normalized residual retains the descended nef moduli part together with the effective klt discriminant and the residual properties needed for the section comparisons below. The parabolic pullback statement and independence of the chosen root trivialization are also given directly in [1, Definition-Proposition 7.1].

Remark 6.2. No abundance assertion for M is part of the input. The stronger conclusion of [2, Theorem 3.3] assumes effectivity of the boundary on the geometric generic fibre, an assumption which need not hold for $-R_C$. The b-nefness and Hodge identification used above instead

come from the results of [1] just cited. In particular, no good minimal model of C is required. For the Hodge-theoretic formulation with sub-pairs, see also [5, Definition 6.18, Construction-Definition 6.23, Remark 6.25, and Theorem 6.28].

6.2. The least-index root cover and its rational variation.

Lemma 6.3 (Canonical root cover). *Let C be a smooth connected projective variety with $\kappa(C) = 0$, and let $b > 0$ be the least integer for which $H^0(C, bK_C) \neq 0$. Choose a nonzero section s of bK_C and a nonzero rational canonical form ω . Let T be the normalization of C in the field obtained by adjoining a root*

$$\xi^b = \frac{s}{\omega^b}.$$

Then $T \rightarrow C$ has degree b , and every smooth projective resolution W of T satisfies

$$\kappa(W) = 0, \quad h^0(W, K_W) = 1.$$

The unique canonical line is spanned by the form $\xi h^ \omega$, where $h : W \rightarrow C$ is the induced morphism.*

Proof. Put $n = \dim C$ and $D_C = \operatorname{div}(s)$. The case $n = 0$ is immediate. For $n > 0$, first note that every nonzero pluricanonical space on C has dimension one: two independent sections in the same degree would give a nonconstant rational map and hence positive Kodaira dimension.

The field $\mathbf{C}(C)$ contains all roots of unity. If $s/\omega^b = u^p$ for a prime $p \mid b$, the rational (b/p) -canonical form $u\omega^{b/p}$ has p -th power s . At every prime divisor its order is p^{-1} times the nonnegative order of s , so this form has no poles. A rational section of a line bundle on the smooth variety C with no divisorial poles is regular. This contradicts the minimality of b . The Kummer irreducibility criterion now gives a field extension of degree b . Denote the normalization map by $\pi : T \rightarrow C$, and a smooth projective resolution by $\rho : W \rightarrow T$, so $h = \pi\rho$.

The differential of h gives a nonzero morphism $h^*K_C \rightarrow K_W$ at the generic point, and a regular morphism of line bundles everywhere. Use this morphism and its tensor powers when pulling back canonical forms. The rational canonical form $\eta = \xi h^* \omega$ then satisfies

$$\eta^{\otimes b} = h^* s.$$

The right side is a regular section of bK_W . Thus $b \operatorname{ord}_E(\eta) \geq 0$ for every prime divisor $E \subset W$. Normality of W again implies that η is regular, and therefore $h^0(W, K_W) \geq 1$.

We next bound the canonical divisor from above. Choose $K_C = \operatorname{div}(\omega)$, and on T use the canonical Weil divisor determined by the differential pullback of ω . At a prime divisor $P \subset C$ outside $\operatorname{Supp} D_C$, the order of s/ω^b is divisible by b . Locally over the discrete valuation ring at P , after dividing the root by a power of a uniformizer, the defining equation is a b -th root of a unit. It is unramified, since the characteristic is zero. Consequently all ramification divisors lie over $\operatorname{Supp} D_C$.

If $Q \subset T$ lies over $P \subset \operatorname{Supp} D_C$, write $e = e(Q/P)$ and $d = \operatorname{ord}_P(D_C) \geq 1$. Tameness gives the ramification coefficient $e - 1$, and

$$e - 1 \leq ed = \operatorname{ord}_Q(\pi^* D_C).$$

The canonical divisor formula for finite maps, interpreted at codimension-one points, therefore gives

$$K_T \leq \pi^*(K_C + D_C).$$

On W the same inequality holds away from the ρ -exceptional divisors. Hence there is an effective integral ρ -exceptional divisor E such that

$$(6.3) \quad K_W \leq h^*(K_C + D_C) + E.$$

This argument does not require a pullback of the Weil divisor K_T .

Set $A = K_C + D_C$. Normality of T implies $\rho_* \mathcal{O}_W(mE) = \mathcal{O}_T$ for every $m \geq 0$: a rational function allowed poles only on exceptional divisors is regular in codimension one on T , hence regular on T . Projection formula and (6.3) imply

$$\kappa(W, K_W) \leq \kappa(W, h^*A + E) = \kappa(T, \pi^*A) = \kappa(C, A).$$

The last equality is invariance of Iitaka dimension under a finite surjective pullback. In this situation it can be seen directly from section rings: a homogeneous section upstairs satisfies its monic characteristic polynomial over $\mathbf{C}(C)$; its coefficients, in degree jm , are sections of jmA , by the valuation bounds for sections of $m\pi^*A$. Thus the upstairs graded ring is integral over the downstairs ring, and their fraction fields have the same transcendence degree. Finally $D_C \sim bK_C$, so

$$\kappa(C, A) = \kappa(C, (b+1)K_C) = 0.$$

Together with the nonzero form η , this proves $\kappa(W) = 0$, and then $h^0(W, K_W) = 1$. $\square$

The underlying cover of Lemma 6.3 is the canonical root cover used in [15, Remark 2.6 and Theorem 3.1]. Over a dense open subset $U \subset Y$, a rational relative pluricanonical section constructs this cover in families; a projective resolution, followed by shrinking U , gives a smooth projective morphism $h_U : \mathcal{W}_U \rightarrow U$. Its geometric generic fibre has the invariants in Lemma 6.3. Constancy of Hodge numbers in the smooth projective family then gives geometric genus one on every fibre. If their dimension is n , then

$$\mathbb{V} = R^n(h_U)_* \mathbf{Q}, \quad \text{rk } F^n(\mathbb{V} \otimes \mathcal{O}_U) = 1.$$

The variation has an integral lattice modulo torsion and is polarizable, using a relative ample class and the Lefschetz decomposition. Alternatively, its primitive middle cohomology is polarizable and contains all of $H^{n,0}$, since cup product with a $(1,1)$ -class annihilates $H^{n,0}$. The cyclic deck transformation $\xi \mapsto \zeta\xi$ acts on the unique top form through ζ . Thus the distinguished canonical eigensheaf is the entire top Hodge line of a rational polarizable variation, even when the character itself is not defined over $\mathbf{Q}$. We do not replace the rational variation by that character eigenspace.

After a smooth projective alteration $\tau : T_Y \rightarrow Y$, resolution of the boundary, and semistable reduction in codimension one, the identification in Theorem 6.1 reads

$$(6.4) \quad \tau^*M \sim_{\mathbf{Q}} aJ, \quad a \in \mathbf{Q}_{>0},$$

where J is the Schmid extension of the top Hodge line of $\tau^*\mathbb{V}$, with unipotent local monodromy. Birational changes of the resolved root cover do not change its canonical direct image: on a common resolution, differential pullback identifies its canonical forms. The extension comparison is the one in [1, Theorem 4.4(1),(3), Lemma 5.2(5), and Proposition 5.5], rather than an identification asserted only on U . Changing the chosen rational trivialization by a function from the base gives the same identification after a further finite base change extracting the corresponding root.

6.3. Relative Iitaka reduction and section comparisons.

Proposition 6.4 (Canonical bundle reduction). *Let $f : X \rightarrow Z$ be an algebraic fibre space between smooth connected projective varieties, and let its geometric generic fibre F have $d = \kappa(F) \geq 0$. There exist smooth projective varieties $\tilde{X}, Y$, a birational morphism $\mu : \tilde{X} \rightarrow X$, and algebraic fibre spaces*

$$\tilde{X} \xrightarrow{g} Y \xrightarrow{q} Z, \quad qg = f\mu,$$

such that $\dim Y - \dim Z = d$ and the geometric generic fibre of g has Kodaira dimension zero. They admit the data B, M, R of Theorem 6.1, with R^- exceptional over X . The moduli divisor M has the realization (6.4) for a polarizable rational variation whose top Hodge piece has rank one. Writing

$$D = K_Y + B + M,$$

one has

$$(6.5) \quad \kappa(X) \geq \kappa(Y, D), \quad D|_{Y_{\bar{\eta}}} \text{ is big,}$$

where $\bar{\eta}$ is a geometric generic point of Z . The same bigness holds on very general smooth fibres of q .

Proof. Over $\mathbf{C}(Z)$, take a sufficiently divisible pluricanonical map of the generic fibre and its Iitaka fibration, including the relative algebraic closure of the image field. The Iitaka fibration theorem gives image dimension d and Kodaira dimension zero for the geometric generic fibre of this fibration. Formation of its pluricanonical spaces commutes with extension to an algebraic closure of $\mathbf{C}(Z)$; thus these statements can be checked geometrically. Extend the rational map to projective models over Z , resolve source and target, and take the connected-fibre factorization. This gives $X_0 \rightarrow Y_0 \rightarrow Z$ with the required dimensions. The second map has connected fibres as well: pushing $\mathcal{O}_{X_0}$ first to Y_0 and then to Z gives $\mathcal{O}_Z$, because the original f and the birational map to its smooth source have this property.

Apply Theorem 6.1 to $X_0 \rightarrow Y_0$. All its modifications can be performed over Z . Exceptionality over X_0 implies exceptionality over X . Lemma 6.3 and the subsequent family construction give the required rational variation and its top line. The two comparisons are proved in Lemma 6.5 below. $\square$

Lemma 6.5 (Comparison of section growth). *For the factorization and canonical bundle data in Proposition 6.4, (6.5) holds.*

Proof. The formula gives

$$K_{\tilde{X}} + R^- \sim_{\mathbf{Q}} g^*D + R^+.$$

Write $K_{\tilde{X}} = \mu^*K_X + E_\mu$, using compatible canonical divisors. Since X and $\tilde{X}$ are smooth, E_μ is effective and μ -exceptional. The same is true of $E_\mu + R^-$. For sufficiently divisible m , normality and projection formula therefore give

$$H^0(\tilde{X}, m(K_{\tilde{X}} + R^-)) \simeq H^0(X, mK_X).$$

Multiplication by the canonical section of mR^+ injects $H^0(Y, mD)$ into the space on the left; here we also use $g_*\mathcal{O}_{\tilde{X}} = \mathcal{O}_Y$. This proves $\kappa(Y, D) \leq \kappa(X)$.

Now make the flat base change $\bar{\eta} \rightarrow Z$. The varieties $\tilde{X}_{\bar{\eta}}$ and $Y_{\bar{\eta}}$ are smooth; the former is birational to the geometric generic fibre F of f . Restrictions of canonical divisors agree, up to linear equivalence, with the corresponding canonical divisors of these fibres. The restricted formula consequently implies

$$K_{\tilde{X}_{\bar{\eta}}} \leq_{\mathbf{Q}} g_{\bar{\eta}}^*(D|_{Y_{\bar{\eta}}}) + R^+|_{\tilde{X}_{\bar{\eta}}}.$$

Here $\leq_{\mathbf{Q}}$ means that the difference is $\mathbf{Q}$ -linearly equivalent to an effective divisor. Formation of a proper direct image commutes with this flat base change, so (6.2) yields, for sufficiently divisible m ,

$$(g_{\bar{\eta}})_*\mathcal{O}_{\tilde{X}_{\bar{\eta}}}(mR^+|_{\tilde{X}_{\bar{\eta}}}) = \mathcal{O}_{Y_{\bar{\eta}}}.$$

Projection formula now gives

$$\begin{aligned} h^0(\tilde{X}_{\bar{\eta}}, mK_{\tilde{X}_{\bar{\eta}}}) &\leq h^0(\tilde{X}_{\bar{\eta}}, mg_{\bar{\eta}}^*(D|_{Y_{\bar{\eta}}}) + mR^+|_{\tilde{X}_{\bar{\eta}}}) \\ &= h^0(Y_{\bar{\eta}}, mD|_{Y_{\bar{\eta}}}). \end{aligned}$$

It follows that

$$d = \kappa(F) \leq \kappa(Y_{\bar{\eta}}, D|_{Y_{\bar{\eta}}}) \leq \dim Y_{\bar{\eta}} = d.$$

Thus $D|_{Y_{\bar{\eta}}}$ is big, including the zero-dimensional case.

For completeness, the bigness assertion spreads to a dense open subset after restricting to the smooth locus of q . Fix a relatively ample divisor H . Bigness on the geometric generic fibre gives, for some sufficiently divisible integer m , a nonzero section of $mD - H$ there. Flat scalar extension of cohomology first gives such a section over $\mathbf{C}(Z)$. Shrinking the base, the section extends to the family and remains nonzero on its fibres. Hence $mD|_{Y_z}$ is the sum of the ample divisor $H|_{Y_z}$ and an effective divisor for general z , proving the stated very-general-fibre assertion. $\square$

Lemma 6.6 (Further birational modifications). *Suppose Y, B, M, D satisfy the conclusions of Proposition 6.4. Let $\nu : Y' \rightarrow Y$ be a smooth projective log resolution, and write*

$$K_{Y'} + B^{\text{cr}} = \nu^*(K_Y + B).$$

*Set $B' = (B^{\text{cr}})^+$, $M' = \nu^*M$, and $D' = K_{Y'} + B' + M'$. Then B' is effective, (Y', B') is klt, M' is nef with the same Hodge realization after pullback, and*

$$D' = \nu^*D + (B^{\text{cr}})^-.$$

The additional divisor is effective and ν -exceptional. Consequently $\kappa(Y', D') = \kappa(Y, D)$, and D' is big on the geometric generic fibre over Z .

Proof. The nonexceptional coefficients of B^{cr} are those of the effective divisor B , so its negative part is exceptional. All coefficients of B^{cr} are less than one by the klt hypothesis. Its positive part has simple normal-crossing support and coefficients in $[0, 1)$, proving that (Y', B') is klt. The displayed identity follows directly from the definitions. Exceptional effective twists do not change section spaces of a pullback to a normal base, so they do not change the Iitaka dimension. On the geometric generic fibre over Z , the induced map is birational and the restricted identity expresses D' as the pullback of a big divisor plus an effective divisor. Nefness is preserved by pullback. For the Hodge assertion, use a common smooth alteration of Y' and the alteration in (6.4); the unipotent Deligne and Schmid extensions commute with the resulting morphisms of smooth normal-crossing pairs. $\square$

Remark 6.7. After Lemma 6.6, B' need not be described as the discriminant of the original subpair. The properties used later are precisely effectiveness and klt singularities, the nef Hodge realization of M' , and the two comparisons in (6.5); all have been preserved. In particular, since the big cone of $Y_{\bar{\eta}}$ is open, there is a rational number v with $0 < v < 1$ such that

$$(K_Y + B + vM)|_{Y_{\bar{\eta}}} \text{ is big.}$$

This remains true on very general fibres by the spreading argument in Lemma 6.5.

7. ORDINARY SUBADDITIVITY AND EXTENSION OF SCALARS

We complete Theorem 1.1. Section 6 has reduced the problem to an effective klt pair (Y, B) with a nef Hodge line M . The period theorem writes $M = p^*L$. The original base map $q : Y \rightarrow Z$ and this new map $p : Y \rightarrow S$ need not factor through one another. Their combined image will first give nonvanishing on a period fibre. Weak positivity and one exact interpolation identity then supply the missing total-space sections.

We first work over $\mathbf{C}$. Two established positivity results will be used in precisely the following forms.

Theorem 7.1 (Subadditivity with log-general-type fibre). *Let $u : V \rightarrow T$ be a surjective morphism with connected fibres, where V is normal projective and T is smooth projective. Let $\Delta \geq 0$ be a $\mathbf{Q}$ -divisor such that (V, Δ) is log canonical. If the log canonical divisor of the geometric generic fibre G is big and $\kappa(T) \geq 0$, then*

$$\kappa(V, K_V + \Delta) \geq \dim G + \kappa(T).$$

This is the log-general-type-fibre theorem of Kovács–Patakfalvi [23, Theorem 9.9], in the normal-total-space form [18, Theorem 2.11], with the auxiliary divisor equal to K_T . It also follows from [18, Theorem 1.4(1)]: a big divisor is abundant. In particular, the theorem does not require abundance of K_T .

Theorem 7.2 (Twisted weak positivity). *Let $u : V \rightarrow T$ be a surjective morphism from a normal projective variety to a smooth projective variety, and let (V, Δ) be an effective log canonical $\mathbf{Q}$ -pair. If k is a positive integer such that $k(K_V + \Delta)$ is Cartier, then*

$$u_* \mathcal{O}_V(nk(K_{V/T} + \Delta))$$

is weakly positive for every positive integer n .

This is the projective case of [14, Theorem 1.1]. The meaning needed here is that, for a torsion-free weakly positive sheaf $\mathcal{E}$, every ample Cartier divisor A , and every positive integer α , there is a positive integer β such that

$$(\mathrm{Sym}^{\alpha\beta} \mathcal{E})^{**} \otimes \mathcal{O}_T(\beta A)$$

is generated by global sections at the generic point; see [14, Definitions 7.1–7.2].

7.1. Nonvanishing on fibres of the period projection.

Lemma 7.3. *Let $p : Y \rightarrow S$ and $q : Y \rightarrow Z$ be algebraic fibre spaces between smooth projective varieties. Let (Y, B) be an effective klt $\mathbf{Q}$ -pair and let $M \sim_{\mathbf{Q}} p^*L$ for a $\mathbf{Q}$ -divisor L on S . Suppose that $\kappa(Z) \geq 0$ and*

$$D := K_Y + B + M$$

is big on the geometric generic fibre of q . Then the geometric generic fibre of p has nonnegative log Kodaira dimension.

Proof. Take the Stein factorization of the map from Y to its image in $S \times Z$. Resolve its intermediate normal variety and resolve the induced rational map from Y , choosing the modification of Y also to be a log resolution of (Y, B) . We obtain morphisms

$$Y \xrightarrow{r} W, \quad t : W \rightarrow S, \quad h : W \rightarrow Z, \quad p = t \circ r, \quad q = h \circ r,$$

where Y, W are smooth projective, r has connected fibres, and $W \rightarrow S \times Z$ is generically finite onto its image. We continue to write Y, B, M, D for the modified data. This replacement is harmless: for such a log resolution $\mu : Y' \rightarrow Y$, take the positive part B' of the crepant boundary. Then

$$K_{Y'} + B' = \mu^*(K_Y + B) + E, \quad E \geq 0$$

with E exceptional, and (Y', B') is klt. The same identity on the general p -fibres preserves their log Kodaira dimensions. Pullback of M and the effective correction preserve the required bigness on the geometric generic q -fibre.

Since $p_* \mathcal{O}_Y = \mathcal{O}_S$ and $r_* \mathcal{O}_Y = \mathcal{O}_W$, the fibres of t are connected. For general $s \in S$, the fibre W_s is smooth and connected, and $h_s : W_s \rightarrow Z$ is generically finite onto its image. Write

$$a = \dim W_s, \quad m = \dim Z.$$

The generically finite map onto its image gives $a \leq m$. We claim that $\kappa(W_s) \geq 0$. The assertion is immediate for $a = 0$; assume $a > 0$. Along a smooth fibre of t , the cotangent sequence is

$$0 \longrightarrow \Omega_{S,s}^1 \otimes \mathcal{O}_{W_s} \longrightarrow \Omega_{W|W_s}^1 \longrightarrow \Omega_{W_s}^1 \longrightarrow 0.$$

Since the last term has rank a , the entire m -th exterior power lies in the filtration term with at least $m - a$ base factors. Its quotient by the next term gives, after pulling back top forms on Z , a morphism

$$(7.1) \quad h_s^* \omega_Z \longrightarrow \bigwedge^{m-a} \Omega_{S,s}^1 \otimes \omega_{W_s}.$$

At a general point of a general W_s , the differential of h has rank m , and its restriction to the tangent space of W_s has rank a . Thus (7.1) is nonzero. A suitable linear functional on the constant exterior-power factor yields a nonzero morphism $h_s^* \omega_Z \rightarrow \omega_{W_s}$.

Choose $n > 0$ and $0 \neq \tau \in H^0(Z, nK_Z)$. For general s , the image $h_s(W_s)$ is not contained in the zero divisor of τ : otherwise dominance of h would fail. The n -th tensor power of the preceding morphism therefore sends $h_s^* \tau$ to a nonzero section of nK_{W_s} . This proves the claim.

Let G be a general fibre of r . For general $z \in Z$, such fibres form a moving family covering Y_z . A big divisor restricts to a big divisor on a general member of this family: restrict an ample-plus-effective decomposition, choosing a member not contained in the support of the effective part. Consequently $D|_G$ is big. Since G maps to a point of S , $M|_G \sim_{\mathbf{Q}} 0$, and adjunction gives

$$D|_G \sim_{\mathbf{Q}} K_G + B|_G.$$

For general s , the morphism $r_s : Y_s \rightarrow W_s$ has connected fibres, $(Y_s, B|_{Y_s})$ is klt, and its general fibre has big log canonical divisor. Theorem 7.1 gives

$$\kappa(Y_s, K_{Y_s} + B|_{Y_s}) \geq \dim G + \kappa(W_s) \geq 0.$$

Finally choose s outside the countably many proper jumping loci for the spaces of sections of all divisible log pluricanonical bundles on the smooth p -fibres. Generic base change identifies their dimensions with those on the geometric generic fibre. Thus the last nonvanishing holds on that fibre as well. $\square$

7.2. Effectivity from weak positivity.

Lemma 7.4. *Let $p : Y \rightarrow S$ be an algebraic fibre space between smooth projective varieties, and let (Y, B) be an effective klt $\mathbf{Q}$ -pair. Suppose that the geometric generic fibre has nonnegative log Kodaira dimension. For every big $\mathbf{Q}$ -divisor P on S , the divisor*

$$K_{Y/S} + B + p^* P$$

is $\mathbf{Q}$ -linearly equivalent to an effective divisor.

Proof. Choose a dense open over which p is flat. Flatten p by a projective birational modification of the base and then resolve the base, retaining flatness of the main transform by base change; see [26, Lemma 38.31.1, Tag 081R]. Resolve the resulting main transform $V \rightarrow S_1$, choosing the projective resolution so that its composite $\mu : Y_1 \rightarrow Y$ is also a log resolution of (Y, B) . Thus

$$\sigma : S_1 \rightarrow S, \quad \mu : Y_1 \rightarrow Y, \quad p_1 : Y_1 \rightarrow S_1$$

are compatible morphisms, S_1, Y_1 are smooth projective, σ, μ are birational, and $Y_1 \rightarrow V$ is a resolution with $V \rightarrow S_1$ flat. Any prime divisor of Y_1 whose image in S_1 has codimension at least two is μ -exceptional. Indeed, flatness implies that the inverse image of such a base subset has codimension at least two in V ; hence the divisor is exceptional over V , and its image in Y also has codimension at least two.

Take the positive part B_1 of the crepant boundary, so that (Y_1, B_1) is effective klt and

$$(7.2) \quad K_{Y_1} + B_1 = \mu^*(K_Y + B) + E, \quad E \geq 0, \quad E \text{ } \mu\text{-exceptional}.$$

Generic-fibre nonvanishing and field-extension base change allow us to choose a sufficiently divisible $k > 0$ for which

$$\mathcal{E} = p_{1*} \mathcal{O}_{Y_1}(k(K_{Y_1/S_1} + B_1))$$

has positive rank. Theorem 7.2 applies to $\mathcal{E}$.

Fix an ample Cartier divisor A on S_1 , and choose a positive integer α so large that

$$\sigma^*P - \frac{1}{k\alpha}A$$

is big. For some $\beta > 0$, weak positivity makes

$$(\mathrm{Sym}^{\alpha\beta} \mathcal{E})^{**} \otimes \mathcal{O}_{S_1}(\beta A)$$

generically generated. Set $\ell = \alpha\beta$. Let $U \subset S_1$ be the locally free locus of the torsion-free sheaf $\mathcal{E}$; its complement has codimension at least two. On all of U , multiplication of sections gives

$$\mathrm{Sym}^\ell \mathcal{E}|_U \longrightarrow p_{1*} \mathcal{O}_{Y_1}(k\ell(K_{Y_1/S_1} + B_1))|_U.$$

This map is nonzero at the generic point: a nonzero section on the integral generic fibre has nonzero powers. Generic generation therefore supplies a global section of the twisted reflexive symmetric power whose product is nonzero. Its image is a regular section over $p_1^{-1}(U)$ of

$$k\ell(K_{Y_1/S_1} + B_1) + \beta p_1^*A.$$

Here U is the locally free locus, not merely a smaller open where the chosen sections generate.

The big divisor

$$k\ell\sigma^*P - \beta A = k\ell\left(\sigma^*P - \frac{1}{k\alpha}A\right)$$

has an effective sufficiently divisible multiple. Taking the same power of the preceding section and multiplying by the pullback of a nonzero section of that multiple gives a nonzero section over $p_1^{-1}(U)$ of a multiple of

$$D_1 := K_{Y_1/S_1} + B_1 + p_1^*\sigma^*P.$$

As a rational section on Y_1 , it has poles only above $S_1 \setminus U$. Thus there are $n > 0$, a rational function $\varphi \neq 0$, and an effective μ -exceptional divisor T such that

$$(7.3) \quad \mathrm{div}(\varphi) + nD_1 + T \geq 0.$$

Choose compatible canonical divisors and put

$$D_0 = K_{Y/S} + B + p^*P, \quad R = K_{S_1} - \sigma^*K_S \geq 0.$$

Equation (7.2) gives the exact identity

$$D_1 = \mu^*D_0 + E - p_1^*R.$$

Push (7.3) forward to the smooth variety Y . Since $\mu_*E = \mu_*T = 0$, we obtain

$$\mathrm{div}(\varphi) + nD_0 \geq n\mu_*p_1^*R \geq 0.$$

This proves rational linear effectivity of D_0 . □

7.3. Completion over the complex numbers.

Proof of Theorem 1.1 over $\mathbf{C}$. If either $\kappa(F)$ or $\kappa(Z)$ is $-\infty$, the assertion follows from the convention in the statement. Assume therefore that

$$d := \kappa(F) \geq 0, \quad \kappa(Z) \geq 0.$$

Proposition 6.4 and Lemma 6.5 provide algebraic fibre spaces

$$\tilde{X} \longrightarrow Y \xrightarrow{q} Z$$

and an effective klt pair (Y, B) , together with a nef $\mathbf{Q}$ -divisor M having the required Hodge realization. Writing $D = K_Y + B + M$, their conclusions include

$$(7.4) \quad \dim(Y/Z) = d, \quad \kappa(X) \geq \kappa(Y, D), \quad D|_{Y_{\tilde{\eta}}} \text{ big},$$

where $Y_{\bar{\eta}}$ is the geometric generic q -fibre.

Apply adjoint positivity, Theorem 1.2, and the compatible modifications of Lemma 6.6. We obtain an algebraic fibre space $p : Y \rightarrow S$ with S smooth projective and

$$M \sim_{\mathbf{Q}} p^*L, \quad L \text{ nef.}$$

If $\dim S = 0$, then $M \sim_{\mathbf{Q}} 0$. Theorem 7.1, applied directly to q , and (7.4) give the result. We may consequently suppose $\dim S > 0$. Adjoint positivity also gives

$$K_S + jL \text{ big for all sufficiently large integers } j.$$

Openness of the big cone on $Y_{\bar{\eta}}$ permits a rational number v with $0 < v < 1$ such that

$$(7.5) \quad (K_Y + B + vM)|_{Y_{\bar{\eta}}} \text{ is big.}$$

Lemma 7.3 shows that the geometric generic p -fibre has nonnegative log Kodaira dimension. Choose a sufficiently large integer $j > 1$ and an ample $\mathbf{Q}$ -divisor H on S such that

$$P_0 := K_S + jL - H$$

is big. Lemma 7.4 gives

$$(7.6) \quad E_1 := K_Y + B + jM - p^*H \sim_{\mathbf{Q}} K_{Y/S} + B + p^*P_0 \sim_{\mathbf{Q}} \text{ an effective divisor.}$$

Choose

$$\lambda = \frac{1-v}{j-v} \in (0, 1) \cap \mathbf{Q}, \quad \lambda j + (1-\lambda)v = 1.$$

This balances the coefficients of M ; to cancel the corresponding term $-\lambda p^*H$, put

$$A_S = vL + \frac{\lambda}{1-\lambda}H, \quad E_2 = K_Y + B + p^*A_S.$$

Then A_S is ample and

$$(7.7) \quad D \sim_{\mathbf{Q}} \lambda E_1 + (1-\lambda)E_2.$$

Choose a sufficiently divisible integer $N > 1$ such that $|Np^*A_S|$ is basepoint-free. A general member $T_N \in |Np^*A_S|$ gives

$$\Gamma = \frac{1}{N}T_N \sim_{\mathbf{Q}} p^*A_S$$

with $(Y, B + \Gamma)$ klt. This follows from Bertini, or after pulling the basepoint-free system to a log resolution of (Y, B) and choosing N sufficiently large. Thus $E_2 \sim_{\mathbf{Q}} K_Y + B + \Gamma$ is the log canonical class of an effective klt pair. Its restriction to $Y_{\bar{\eta}}$ is the big divisor in (7.5) plus the nef divisor $\lambda p^*H/(1-\lambda)$, and hence is big. Theorem 7.1 now yields

$$\kappa(Y, E_2) \geq d + \kappa(Z).$$

In sufficiently divisible degrees, multiplication by powers of a nonzero section in (7.6) embeds the corresponding linear systems of $(1-\lambda)E_2$ into those of D . Positive rational scaling leaves Iitaka dimension unchanged, so

$$\kappa(X) \geq \kappa(Y, D) \geq \kappa(Y, E_2) \geq \kappa(F) + \kappa(Z),$$

as required. □

7.4. Extension to arbitrary characteristic-zero fields.

Lemma 7.5. *Let $K \subset K'$ be a field extension and let V be a smooth projective geometrically integral variety over K . The Kodaira dimension of V is unchanged by extending scalars to K' . The same assertion holds for the geometric generic fibre of an algebraic fibre space between smooth projective geometrically integral varieties when the ground field is extended.*

Proof. Flat base change for proper coherent cohomology gives, in every positive degree n ,

$$H^0(V, nK_V) \otimes_K K' \simeq H^0(V_{K'}, nK_{V_{K'}}).$$

The canonical bundle commutes with this smooth scalar extension. Nonvanishing is therefore unchanged. When the space is nonzero, choose a K -basis. The rational map defined by that basis and its image base-change to the complete pluricanonical map over K' ; dimensions of finite-type schemes are unchanged by extension of the ground field. Taking the supremum of these image dimensions proves the first assertion, including the case $-\infty$.

For the fibre assertion, put $E = K(Z)$ and $E' = K'(Z_{K'})$, and write F_K for the generic fibre over E . The generic fibre after ground-field extension is $F_K \times_E E'$. Choose an algebraic closure Ω of E' and an algebraic closure $\bar{E} \subset \Omega$ of E . The two geometric generic fibres are $F_K \times_E \bar{E}$ and $F_K \times_E \Omega$; the latter is a scalar extension of the former. Their Kodaira dimensions agree by the first assertion. $\square$

To complete the proof of Theorem 1.1, let k be any algebraically closed field of characteristic zero. The varieties, their projective embeddings, and the morphism descend to a finitely generated subfield $K \subset k$ over $\mathbf{Q}$. Smoothness, geometric integrality and surjectivity hold for the descended data by faithful flatness. Proper flat base change and faithful flatness also preserve the equality $f_*\mathcal{O}_X = \mathcal{O}_Z$, hence the connected-fibre condition. Choose an embedding $K \hookrightarrow \mathbf{C}$. Lemma 7.5 identifies all three Kodaira dimensions in the statement with their counterparts over $\mathbf{C}$, using the geometric generic fibre for F . The complex case just proved therefore establishes the theorem over k .

8. AN INTEGRAL PROJECTIVE PROOF OF ADJOINT POSITIVITY

We give another proof of adjoint positivity for the whole highest Hodge line of a rationally polarized integral pure variation. The conclusion is the same adjoint conclusion as Theorem 1.2 for the rational parabolic extension itself. A positive rational multiple has the same conclusion after rescaling the large adjoint parameter. We first replace the variation by an integral tensor construction, then descend it through its full period map and subtract the boundary by comparing intersection growth.

The tensor construction removes every monodromy element acting trivially on the moving projective line. This makes a transverse loop detect strict loss of the line's numerical dimension at each remaining boundary divisor. The full period map supplies an algebraic base; the differential of the map remembering only the highest line controls the boundary subtraction. These two ranks need not agree.

Let P be a smooth connected projective complex variety, let D be a reduced simple normal crossing divisor, and put $P^\circ = P \setminus D$. A *top Hodge line* of a rationally polarized integral pure variation $\mathbb{V}$ on P° means

$$F^p \mathbb{V}_\mathcal{O} \quad \text{with} \quad F^{p+1} \mathbb{V}_\mathcal{O} = 0, \quad \text{rank } F^p \mathbb{V}_\mathcal{O} = 1.$$

Here an integral variation has a local system of finite-rank free $\mathbb{Z}$ -modules, and the polarization is rational and flat. The weight and p may be arbitrary integers. The line itself is a complex

Hodge subbundle; no rationality assumption is made on it. If V is a flat reference fiber, its *line period map* is the equivariant holomorphic map

$$\ell : \widetilde{P}^\circ \longrightarrow \mathbb{P}(V), \quad x \longmapsto F_x^p.$$

Its generic differential rank does not depend on the reference fiber.

Theorem 8.1 (Adjoint positivity). *Let P be a smooth connected projective complex variety and D a reduced simple normal crossing divisor. Let $\mathbb{V}$ be a rationally polarized integral pure variation on $P \setminus D$ whose highest nonzero Hodge filtration piece is a line, and let $L \in \text{Pic}(P) \otimes \mathbf{Q}$ be its rational parabolic extension. There are a projective birational morphism $\pi : \widetilde{P} \rightarrow P$, with $\widetilde{P}$ smooth, a smooth projective variety Z , a surjective morphism $h : \widetilde{P} \rightarrow Z$ with connected fibers, and a nef rational divisor A on Z such that*

$$\pi^*L \sim_{\mathbf{Q}} h^*A, \quad K_Z + aA \text{ is big for every sufficiently large rational } a.$$

Here a divisor on a point is big. Further smooth birational modifications of the source are permitted by replacing h with its composition with the modification.

We first specify the extension and metric facts used in the proof.

8.1. Extensions and numerical dimension. The monodromy theorem makes the local monodromies quasi-unipotent. For unipotent local monodromy, L is the top filtration subbundle in the Deligne–Schmid extension. For quasi-unipotent monodromy, its rational extension is characterized by the following rule: after an adapted cover making local monodromy unipotent, its pullback is the unipotent extension, as a rational line bundle. This convention includes the parabolic weights; it is not an arbitrary choice of a Deligne lattice.

Lemma 8.2 (Extension facts). *The rational extension has the following properties.*

- (i) *It is nef and is compatible with positive tensor powers and pullback by morphisms of smooth projective log pairs whose interiors map to the original interior.*
- (ii) *In the unipotent case, on each connected boundary stratum the local limiting mixed Hodge structures determine a unique weight-graded piece containing the top Hodge line. These pieces glue to a polarizable pure variation with rank-one top filtration, whose extended top line on the closure of the stratum is the restriction of L .*
- (iii) *The preceding restriction statement remains valid after an adapted cover in the quasi-unipotent case. In particular, numerical dimensions of restrictions may be computed on a smooth adapted cover of the stratum.*

Proof. In the unipotent case the canonical extension and relative monodromy weight filtration are supplied by nilpotent-orbit theory [25, 11]; precise formulations for the top line are given in [5, §2.5.3, Lemma 2.16, Lemma 5.6]. The graded pieces are polarized by the primitive decomposition.

For a boundary stratum, take the weight filtration associated with an interior point of its monodromy cone; it is independent of that choice. Strictness and rank $F^p = 1$ select one nonzero highest graded piece. The global object in (ii) is this pure graded variation. To check its gluing when the normal bundle is nontrivial, choose a local normal coordinate t and write N for its monodromy logarithm. Replacing t by $t' = u(s)t$, with u invertible along the stratum, changes the local limiting filtration by

$$\exp\left(-\frac{\log u(s)}{2\pi\sqrt{-1}}N\right).$$

Since N lowers the monodromy weight filtration by two, this transition acts identically on each weight graded. For several normal coordinates the same calculation uses the sum of the commuting monodromy logarithms, which also acts trivially on their common weight graded. The pure graded data therefore glue. Strictness identifies the rank-one top filtration with its unique

nonzero top graded piece. At the remaining corners, [5, §2.5.3 and Lemma 2.16(3)] identifies the canonical extensions on the entire smooth stratum closure, so the identification of lines holds there as well.

For completeness, the rational convention is compatible with these operations as follows. In a crossing chart, take sufficiently divisible roots of its boundary coordinates. The resulting unipotent extension is equivariant for the finite deck group. A common positive power kills the characters of the stabilizers on its line fibers and descends to a line bundle in the original chart. The descents agree on overlaps: compare them on a common root cover and use the uniqueness of the unipotent extension. This defines a class in $\text{Pic}(P) \otimes \mathbf{Q}$ and proves compatibility with further pullbacks and powers. One can perform the comparison globally on a smooth adapted alteration; auxiliary branch divisors cause no difficulty because the original variation extends over them. Nefness descends under a proper surjective morphism, since every curve downstairs is dominated by a curve upstairs. Finally, nef numerical dimension is unchanged by a generically finite pullback, so the same construction on strata proves (iii). $\square$

Lemma 8.3 (Curvature and intersections). *The numerical dimension of the rational top Hodge line equals the generic rank of its line period map:*

$$\nu(L) = \text{rank}(d\ell)_{\text{gen}}.$$

In the unipotent case, if $d = \dim P$, ω is a smooth Kähler form, and α is the first Chern form of the Hodge metric on $L|_{P^\circ}$, then, for $0 \leq k \leq d$,

$$(8.1) \quad \int_{P^\circ} \alpha^k \wedge \omega^{d-k} = c_1(L)^k \cdot [\omega]^{d-k}.$$

The integrals are absolutely convergent.

Proof. The closed-test-form version of this calculation is proved in Lemma 9.1; that proof uses only the extension facts in Lemma 8.2, not any descent or rank-drop result. Taking the test form to be ω^{d-k} gives Equation (8.1). Griffiths' curvature formula gives

$$(8.2) \quad \ker \alpha = \ker(d\ell).$$

If r is the generic differential rank, then $\alpha^{r+1} = 0$ everywhere on the open set, whereas $\alpha^r \wedge \omega^{d-r}$ is positive on a nonempty open subset. The intersection identity and nefness therefore give $\nu(L) = r$. For quasi-unipotent monodromy, both numerical dimension and differential rank are unchanged by a smooth adapted generically finite cover, and Lemma 8.2 identifies the pulled-back rational line with its unipotent extension. $\square$

8.2. Eliminating the projective monodromy kernel. For a variation on a connected smooth algebraic variety, we will use the condition

$$(8.3) \quad \gamma\ell(x) = \ell(x) \text{ for every } x \implies \gamma = \text{id} \quad (\gamma \text{ in the monodromy image}).$$

This is a condition on the image of the moving line, rather than on all of the ambient projective space.

Lemma 8.4 (An integral tensor replacement). *Let U be a smooth connected complex algebraic variety, and let $\mathbb{V}$ be a rationally polarized integral pure variation on U whose highest nonzero Hodge filtration piece is a line L° . There are an integer $b > 0$ and a rationally polarized integral pure variation $\mathbb{V}_1$ on U satisfying Condition (8.3), whose highest nonzero Hodge filtration piece is $(L^\circ)^{\otimes b}$. On any smooth projective simple-normal-crossing compactification of U , the rational parabolic highest line of $\mathbb{V}_1$ is the b th power of that of $\mathbb{V}$.*

Proof. For a general algebraic U , choose a dense quasi-projective open U_0 . The map $\pi_1(U_0) \rightarrow \pi_1(U)$ is surjective, since loops can be moved off the complement. Thus the monodromy decompositions below are unchanged, and a flat projector of Hodge type $(0,0)$ on U_0 has that type throughout U by continuity. Write Γ for the original monodromy image. By semisimplicity [13, Theorem 4.2.6 and the added-in-proof note], the complex local system is a sum of monodromy-isotypic summands V_j . These are complex Hodge subvariations. To see this precisely, the fixed-part theorem equips $\text{End}_\Gamma(V)$ with its Hodge algebra structure, and its center has type $(0,0)$ by [13, Corollary 4.2.8(i),(ii)]. Equivalently, the connected Hodge action fixes every primitive central idempotent of this semisimple algebra. The images of those idempotents are exactly the V_j .

The top line belongs to one summand V_{j_0} . Retain that summand and all its rational conjugates. Their sum is a rational subvariation: the central idempotents are defined over a finite splitting field of the rational commutant algebra, and the selected sum is Galois invariant. It inherits the rational polarization and the lattice obtained by intersecting with the original lattice. Denote it by V_0 . None of this construction requires the highest line, or its scalar character, to be rational.

Suppose $\gamma \in \Gamma$ fixes the moving line projectively everywhere. Its projective fixed locus is the finite union of the projectivizations of its eigenspaces. The connected line image lies in one of them, so γ is scalar on the linear span S of that image. Equivariance makes S a nonzero Γ -subrepresentation of V_{j_0} . Write $V_{j_0} = E \otimes M$, with E irreducible and Γ acting trivially on M . Since S contains a copy of E , scalar action on S implies the same scalar action on E and hence on all of V_{j_0} . Rational conjugation gives scalar action on every retained V_j .

It follows that $\gamma|_{V_0}$ preserves the Hodge filtration at any point. It also preserves the rational polarization and the lattice. The stabilizer of a polarized Hodge structure in its real polarization group is compact, since it preserves the positive definite Hodge metric. Its intersection with the integral automorphism group is finite. Therefore $\gamma|_{V_0}$ has finite order. For this particular variation, one may take the exponent of that finite stabilizer at a fixed base point. There is also a choice depending only on $R = \text{rank } V_0$. Indeed, an eigenvalue of an integral finite-order matrix of rank R has order q with $\varphi(q) \leq R$, because its cyclotomic minimal polynomial divides the characteristic polynomial. The finite set of such q therefore gives the explicit permissible choice

$$b = \text{lcm}\{q \geq 1 : \varphi(q) \leq R\}.$$

Every scalar on every retained summand has b th power one. This uniform exponent is chosen before constructing the new variation.

Inside $V_0^{\otimes b}$ take

$$V_1 = \bigoplus_{j \text{ retained}} V_j^{\otimes b}.$$

The sum is taken inside $V_0^{\otimes b}$. The individual terms are distinct direct summands in the tensor expansion, so the sum is direct. It is Galois invariant and hence rational, and each summand is a Hodge subvariation. Restricting the tensor polarization and intersecting with the tensor lattice make it rationally polarized and integral. The summand $V_{j_0}^{\otimes b}$ has top line $(L^\circ)^{\otimes b}$; all other summands have strictly smaller highest Hodge index. Thus this is the unique top line of V_1 .

Every element considered above acts trivially on V_1 . Conversely, if an element of the new monodromy image fixes its moving top line, choose a lift in Γ . Injectivity of the Veronese map $\mathbb{P}(V_{j_0}) \rightarrow \mathbb{P}(V_{j_0}^{\otimes b})$ shows that the lift fixes the original moving line. It is therefore killed on V_1 . This proves Condition (8.3). Compatibility of extensions with tensor powers finishes the proof. $\square$

We now descend the variation itself, using faithfulness to kill monodromy along connected period fibres. Consequently we may normalize the period image in the relative algebraic closure of its function field in $\mathbf{C}(P)$ before estimating boundary ranks. The finite-inertia contradiction in Section 4 required the actual image; here the tensor replacement makes fibre monodromy trivial once it fixes the moving line on an open set.

Lemma 8.5 (Full-period descent). *After applying Lemma 8.4 and making smooth birational modifications of the projective models, the resulting variation descends over a dense open subset of a smooth projective variety Z . Its descended full period map is generically immersive and satisfies Condition (8.3). If A_1 is its rational top Hodge extension, then the original extension satisfies*

$$b\pi^*L \sim_{\mathbf{Q}} h^*A_1$$

for a morphism $h : \tilde{P} \rightarrow Z$ with connected fibers.

Proof. Let $\mathcal{D}$ be the period domain for the chosen polarized lattice, and choose an arithmetic group Γ_{ar} containing the actual monodromy image Γ . By [4, Theorem 1.1], the full period map factors through an algebraic dominant morphism $P^\circ \rightarrow Y_0$, where Y_0 is an irreducible quasi-projective variety and $Y_0^{\text{an}} \rightarrow \Gamma_{\text{ar}} \backslash \mathcal{D}$ is a closed analytic immersion. Normalize Y_0 in the relative algebraic closure of $\mathbf{C}(Y_0)$ in $\mathbf{C}(P)$. This extension is finite, so the normalization is finite over Y_0 . Its function field is relatively algebraically closed in $\mathbf{C}(P)$; in characteristic zero this makes the generic fiber geometrically connected. Shrink source and target so that the latter is smooth and the map is a smooth topologically locally trivial fibration with connected fibers. To obtain this shrinking for the possibly nonproper source, stratify a proper algebraic compactification of the map together with its source boundary and restrict over the open base stratum.

A local lift of the full period map along a fiber takes values in one Γ_{ar} -orbit. These orbits are discrete, including when the quotient has finite stabilizers, so that lift is locally constant in fiber directions. Continuation around a fiber loop therefore fixes the lifted full period. Transport the loop to nearby fibers using the local trivialization. In flat coordinates its actual monodromy fixes the top line on a nonempty analytic open subset of the source. This identity extends to its connected universal cover, and Lemma 8.4 kills that monodromy. The homotopy sequence of the fibration now factors the integral representation through the ordinary fundamental group of the base. The filtration descends as well: on a simply connected base chart the pulled-back flat bundle is trivial, the filtration is constant on each connected fiber, and holomorphic local sections of the smooth map show that the descended filtration is holomorphic. Its transversality and polarization follow by pullback.

The descended period map has the dimension of the original full period image and is generically immersive. Base loops lift to the source, so the same argument and Condition (8.3) show that the descended variation retains that condition. Shrinking introduces no change to it: any monodromy element fixing the line on the smaller open fixes it identically by analytic continuation.

Choose a smooth projective compactification of the base, resolve its boundary, and resolve the rational map from P to this compactification. The resulting morphism $h : \tilde{P} \rightarrow Z$ has connected generic fiber and hence connected fibers, by Stein factorization and normality of Z . Enlarge the source boundary to contain the inverse image of the chosen base boundary and resolve it. The rational extension of the original variation is unchanged where it already extended. Lemma 8.2 identifies its b -th power with h^*A_1 . $\square$

8.3. A strict rank drop at nontrivial monodromy.

Lemma 8.6 (A transverse-loop criterion). *Let $U = \Delta \times \Delta^{d-1}$, with coordinates (z, s) , and let $U^* = \Delta^* \times \Delta^{d-1}$. Suppose that a holomorphic map $\ell : \tilde{U}^* \rightarrow \mathbb{P}(V)$ is equivariant for a linear transformation $T \in \text{GL}(V)$ under one positive turn around $z = 0$, and that $\text{rank}(d\ell) \leq r$. Suppose*

that a holomorphic map $q : U \rightarrow \mathbb{C}^r$ has tangential differential of rank r at a point of $z = 0$, and that on U^* its lift satisfies $\ker(d\ell) \subseteq \ker(dq)$. Then T fixes every line in the image of ℓ on a nonempty open subset. If ℓ is the restriction of a line period map on a connected universal cover, T fixes that entire line image. This includes $r = 0$, when q is the map to a point.

Proof. After shrinking around the given point, the submersion theorem provides coordinates

$$(z, q_1, \dots, q_r, u_1, \dots, u_{d-r-1}).$$

The coordinate z is retained because dq already has rank r on its zero divisor. On the punctured neighborhood, the kernel inclusion and the rank bound imply $\text{rank}(d\ell) = r$ and $\ker(dq) = \ker(d\ell)$. Fix q and u and make one positive turn in the z coordinate. Along every lift of that loop, the derivative of ℓ vanishes, so its value is constant. Its endpoints are related by T , which consequently fixes the initial line. Varying the fixed coordinates proves the claim on an open subset. The projective equality $T\ell = \ell$ then extends by the identity theorem to the connected universal cover. For $r = 0$, the rank bound makes ℓ locally constant and the same loop argument applies with no q coordinates. $\square$

Lemma 8.7 (Boundary rank drop). *Let Z be smooth projective, let D be a reduced simple normal crossing divisor, and let a polarizable integral pure variation on $Z \setminus D$ have a top Hodge line satisfying Condition (8.3). Write A_1 for its rational extension and $r = \nu(A_1)$. If D_i is a component whose transverse monodromy is not the identity, then*

$$\nu(A_1|_{D_i}) < r.$$

Proof. If the boundary rank did not drop, suitable single-valued functions of the moving line would have the same differential kernel as that line. A transverse loop in one of their fibers would then force the monodromy to fix the moving line on an open set, contradicting Condition (8.3) after analytic continuation.

Work at a general point of D_i , with coordinates (z, s) on $\Delta \times \Delta^{d-1}$ and $D_i = \{z = 0\}$. Write $T = T_s T_u$ for the Jordan decomposition of the transverse monodromy and put $N = \log T_u$. Choose $e > 0$ with $T_s^e = 1$ and set $z = t^e$. In flat coordinates the nilpotent orbit extension supplies a holomorphic line map Ψ on the root polydisk such that

$$(8.4) \quad \ell(t, s) = \exp\left(\frac{e \log t}{2\pi\sqrt{-1}} N\right) \Psi(t, s), \quad \Psi(\zeta t, s) = T_s \Psi(t, s), \quad \zeta = \exp(2\pi\sqrt{-1}/e).$$

These are equalities of lines; ℓ is evaluated on the logarithmic cover when necessary.

Let $W = W(N)$, centered at the weight of the variation. On the boundary there is a unique index k in which the top line has a nonzero weight-graded image. The map N is a strict morphism of limiting mixed Hodge structures of type $(-1, -1)$. Since $F^{p+1} = 0$, it gives

$$(8.5) \quad F^p \cap NV = N(F^{p+1}) = 0.$$

Thus the limiting line survives in $C = V/NV$.

We also need to preserve its derivative rank. The primitive decomposition of the fixed graded local system identifies

$$\text{Gr}_k^W V = P_k \oplus N \text{Gr}_{k+2}^W V, \quad P_k \xrightarrow{\sim} \text{Gr}_k^W C.$$

Here k is at least the original weight: lower graded pieces are entirely in the image of N and cannot contain the top line. The second summand has zero F^p , so the graded moving top line lies in the flat primitive summand P_k . Its projective derivative rank is therefore unchanged by the displayed isomorphism. Projecting the boundary line in C further to its relevant graded piece recovers precisely this pure line period map. By Lemmas 8.2 and 8.3, its rank is

$$r_i = \nu(A_1|_{D_i}).$$

This equality may be checked on a global adapted cover: restriction of the pulled-back line to a component dominating D_i is its generically finite pullback, which preserves numerical dimension.

Equation (8.4) makes $\Psi(0, s)$ an eigenline of T_s . Its eigenvalue λ is constant on a sufficiently small connected boundary neighborhood. Let $E = C_\lambda$ and let $q_0 : V \rightarrow E$ be the quotient followed by the eigenspace projection. The projective map

$$Q(t, s) = \mathbb{P}(q_0)(\Psi(t, s))$$

is defined near the chosen boundary point and has boundary rank at least r_i . Since N vanishes on C , it also equals $\mathbb{P}(q_0)(\ell(t, s))$. Moreover T_s acts by a scalar on E , so $Q(\zeta t, s) = Q(t, s)$. Hence Q descends holomorphically to the original (z, s) -polydisk, including $z = 0$. This descent uses a projective map; a choice of a fractional scalar frame is unnecessary.

Suppose $r_i \geq r$. Choose r affine coordinates of Q , denoted $q_1, \dots, q_r$, whose differentials are independent along D_i . The map $q = (q_1, \dots, q_r)$ factors locally through ℓ on the punctured chart. Its differential therefore satisfies the kernel inclusion in Lemma 8.6. The rank bound for ℓ is r by Lemma 8.3. The transverse-loop lemma shows that T fixes the moving highest line everywhere on the connected universal cover. Condition (8.3) gives $T = \text{id}$, contrary to the choice of the divisor. If $r = 0$, the same lemma applies with no coordinate functions. If $r > d - 1$, the supposed inequality $r_i \geq r$ is already impossible. $\square$

8.4. Removing the boundary from the adjoint class.

Lemma 8.8 (A boundary subtraction estimate). *Let Z be smooth projective of positive dimension d , let A_1 be a nef rational divisor, and let $J = \sum_i D_i$ be reduced with smooth components. Suppose $K_Z + J$ is big and*

$$\nu(A_1|_{D_i}) < r := \nu(A_1) \quad \text{for every } i.$$

Then $K_Z + aA_1$ is big for every sufficiently large rational a .

Proof. If $r = 0$, the restriction hypothesis forces $J = 0$, so K_Z is already big. Suppose $r > 0$. By Kodaira's lemma write

$$K_Z + J \sim_{\mathbf{Q}} H + E, \quad H \text{ ample}, \quad E \geq 0,$$

and set $B_a = H + aA_1$ for rational $a \geq 0$. Choose one ample integral divisor C_0 such that both C_0 and $C_0 + J$ are globally generated, and put $H' = H + C_0$. This choice is independent of a and of all section degrees below.

For every sufficiently divisible $m > 0$, filter by kJ for $0 \leq k < m$. The restriction sequences give

$$h^0(Z, m(B_a - J)) \geq h^0(Z, mB_a) - \sum_{k=0}^{m-1} h^0(J, (mB_a - kJ)|_J).$$

Since J is reduced, its structure sheaf injects into $\bigoplus_i \mathcal{O}_{D_i}$. Also

$$mC_0 + kJ = (m - k)C_0 + k(C_0 + J)$$

is globally generated. On each D_i , multiplication by a section which is not identically zero bounds the restricted summand by $h^0(D_i, m(H' + aA_1)|_{D_i})$. We obtain the exact estimate

$$(8.6) \quad h^0(Z, m(B_a - J)) \geq h^0(Z, mB_a) - m \sum_i h^0(D_i, m(H' + aA_1)|_{D_i}).$$

Fix a and divide by $m^d/d!$. Asymptotic Riemann–Roch for the ample divisors involved, along divisible $m \rightarrow \infty$, yields

$$(8.7) \quad \text{vol}(B_a - J) \geq B_a^d - d \sum_i (H' + aA_1)^{d-1} \cdot D_i.$$

No uniform Riemann–Roch remainder in a is needed: the exact inequality in Equation (8.6) holds for every a , and the limit is taken with a fixed.

The first polynomial on the right of Equation (8.7) has degree r , with positive leading coefficient $\binom{d}{r} A_1^r H^{d-r}$. Every restriction polynomial has degree at most $\nu(A_1|_{D_i}) \leq r - 1$. The right side is therefore positive for every sufficiently large a . Finally $K_Z + aA_1 \sim_{\mathbf{Q}} (B_a - J) + E$ is big. $\square$

Proof of Theorem 8.1. Apply Lemmas 8.4 and 8.5. We obtain a variation on a dense open of a smooth projective Z satisfying Condition (8.3), with generically immersive full period map, and rational Hodge extension A_1 satisfying $b\pi^*L \sim_{\mathbf{Q}} h^*A_1$. Resolve the boundary so that it is simple normal crossing. Pullback compatibility retains the line identity. Delete every boundary component with identity transverse monodromy, and call the remaining reduced divisor J .

The variation extends to $Z \setminus J$. Indeed, at a point lying only on deleted components all local monodromies are the identity, and the nilpotent orbit extension with all monodromy logarithms zero is a pure polarized extension. The full period map is still generically immersive. Thus [7, Theorem 1.1 and Remark 1.2(1)] applies to the polarized variation on $Z \setminus J$ and gives the bigness of $K_Z + J$; its hypothesis is immersivity at one point, and it imposes no extra condition on boundary monodromy. Lemma 8.7 applies to every component of J , and Lemma 8.8 proves that $K_Z + aA_1$ is big for all sufficiently large rational a . When $\dim Z = 0$ the assertion holds by convention. Set $A = A_1/b$. This is nef, satisfies $\pi^*L \sim_{\mathbf{Q}} h^*A$, and has the same asserted adjoint positivity after rescaling the large parameter. $\square$

Remark 8.9. The construction permits $\nu(A) < \dim Z$. In particular, the algebraic map h is obtained from the full period map of an integral tensor replacement; its fibers are not asserted to be the leaves of the line period map. Neither abundance nor semiampleness of A occurs in the argument.

9. CHERN INTERSECTIONS AND BOUNDARY LIMITS OF A HIGHEST HODGE LINE

We prove the Chern-intersection formula deferred in Lemma 8.3, and then give a different proof of the boundary restriction estimate in Lemma 8.7. Its two ingredients are a Chern-intersection formula with arbitrary smooth closed test forms and a family of Thom forms concentrating on a boundary component. The limiting Hodge norm computes the tangential curvature at a general boundary point; a product Poincaré bound controls the crossings and permits passage to the restriction intersection number.

Throughout this section, R is a smooth connected projective complex variety, D_R is a reduced simple-normal-crossing divisor, and $\mathbb{V}$ is a polarizable integral pure variation on $R \setminus D_R$. A highest Hodge line is its highest nonzero filtration piece F^p , with $F^{p+1} = 0$ and $\text{rank } F^p = 1$. We denote its rational parabolic extension by A . The parabolic convention includes finite-monodromy weights. Its line period map is the map from the universal cover of $R \setminus D_R$ to the projective space of a fixed flat reference fiber which remembers F^p . Its rank need not equal the full-period rank. For $n = \dim R$ we use

$$\nu(A) = \max\{j \in \{0, \dots, n\} : c_1(A)^j \cdot H^{n-j} > 0\},$$

where H is any ample class and A is nef. In particular, this numerical dimension is zero on a point. Normalize $dd^c = \frac{i}{2\pi} \partial\bar{\partial}$.

Lemma 8.2 supplies nefness and compatibility of the rational extension with positive tensor powers and log-pair pullbacks whose interiors map into the original interior. The Chern-intersection formula below uses these extension properties, but neither full-period descent nor a boundary rank-drop result. For strict rank loss we also use the integral tensor replacement of Lemma 8.4 and the transverse-loop criterion of Lemma 8.6. The replacement is faithful on the moving highest line: a monodromy element fixing that line at every point of the connected

universal cover is the identity. The Thom-form calculation computes restriction intersections from tangential limiting curvature. It uses the local graded Hodge data, without identifying the boundary restriction with a globally extended graded line as in Section 8.

9.1. Curvature and intersection powers.

Lemma 9.1 (Chern intersections with closed test forms). *Let A be the rational parabolic highest line of an integral polarized pure variation on the complement of a reduced simple-normal-crossing divisor in a smooth projective variety R . Then*

$$(9.1) \quad \nu(A) = \text{generic differential rank of its projective line map}$$

provided A is nef. In the unipotent case, if α is the Chern form of its Hodge metric on $R \setminus D_R$, then for every $0 \leq j \leq n = \dim R$ and every smooth closed form γ of type $(n-j, n-j)$, the integral $\int_{R \setminus D_R} \alpha^j \wedge \gamma$ is absolutely convergent and equals $c_1(A)^j \cdot [\gamma]$.

Proof. We may compute after a generically finite smooth alteration on which all boundary monodromies are unipotent. To obtain one, pass to a finite level cover with neat monodromy, compactify, and resolve. Quasi-unipotence then gives unipotence, including for the commuting local monodromies on the resolved boundary. The parabolic pullback property of Lemma 8.2 identifies the new line with the pullback of A . Generically finite pullback preserves nef numerical dimension by the intersection and projection formulas. It also preserves the generic differential rank in characteristic zero. We therefore work on the unipotent model, retaining the notation R, A .

Let α be the semipositive Chern form of the highest-line Hodge metric on the open part. Griffiths' curvature formula [16, Theorem 5.2, Formula 5.3, and Formula 6.18] says that $\alpha(v, \bar{v})$ is a positive fixed multiple of the squared norm of the projective highest-line differential in direction v . In particular, its rank is the rank in (9.1).

Consider a crossing chart with coordinates $z_1, \dots, z_n$ and boundary $z_1 \cdots z_c = 0$. Put $\rho_i = -\log |z_i|^2$ and shrink the chart so that $\rho_i > e$. The horizontal Schwarz estimate, applied separately to the coordinate disks, gives

$$(9.2) \quad 0 \leq \alpha \leq C\Omega, \quad \Omega = \sum_{i=1}^c \frac{i dz_i \wedge d\bar{z}_i}{|z_i|^2 \rho_i^2} + \sum_{i=c+1}^n i dz_i \wedge d\bar{z}_i.$$

Here and below harmless positive normalization constants are absorbed into C . The coordinate estimate follows from the curvature of the horizontal period metric and the singular Ahlfors–Schwarz lemma; see [7, §2.1, Proposition 2.4 and Lemma 2.5]. Positivity and the Hermitian-form Cauchy–Schwarz inequality control the off-diagonal entries and give the uniform product estimate (9.2), even where the period differential is degenerate.

By the nilpotent-orbit theorem [25, Theorem 4.12], a nonvanishing top extension frame, written in flat coordinates on a bounded angular sector, is

$$(9.3) \quad v(z) = \exp \left(\sum_{i=1}^c \frac{\log z_i}{2\pi i} N_i \right) u(z),$$

where the N_i commute and u is holomorphic and nonvanishing. If β is the curvature of a fixed smooth metric on A , write $\alpha - \beta = dd^c \varphi$. The ratio of the two metrics satisfies

$$(9.4) \quad |\varphi(z)| \leq C \left(1 + \sum_{i=1}^c \log \rho_i \right).$$

This is the logarithmic growth estimate for canonical Hodge frames; see [10, Theorem 5.1]. Here the Schwarz estimate also gives a direct proof of the bound needed for integration. The differential of the logarithmic Hodge norm of any flat vector is bounded by a constant times the invariant

period metric, with a constant independent of the point by homogeneity. Travel from a fixed point along coordinate paths in a bounded angular sector. Their product-Poincaré length is bounded by $C(1 + \sum_i \log \rho_i)$, so the Hodge metric and its inverse in a flat frame grow at most as products of powers of the ρ_i . The exponential in (9.3) and its inverse are polynomials in the logarithms, since the N_i are commuting nilpotent operators. Finally, u has ordinary norm bounded above and below on a smaller chart. These three observations give the upper and lower logarithmic bounds in (9.4). We have therefore obtained the two controls needed for the cutoff argument:

$$(9.5) \quad 0 \leq \alpha \leq C\Omega, \quad |\varphi| \leq C\left(1 + \sum_i \log \rho_i\right).$$

Chern-current representation for canonical Hodge extensions is established in [10, Theorem 5.2]. We now justify the highest-line intersection formula with closed test forms, including the absence of boundary contributions. For $0 \leq j \leq n$ and every smooth closed form γ of type $(n-j, n-j)$,

$$(9.6) \quad \int_{R \setminus D_R} \alpha^j \wedge \gamma = c_1(A)^j \cdot [\gamma].$$

The case $j = 0$ is immediate, so assume $j > 0$. Choose a smooth function χ that is one on $(-\infty, 1]$ and zero on $[2, \infty)$, and use products of $\chi((\log \rho_i)/M)$ to cut off the boundary. The ρ_i can be globalized using smooth metrics on the boundary divisor bundles. Denote the resulting cutoff by χ_M . Its second derivatives are uniformly bounded in the product-Poincaré metric, and their support escapes to the boundary as $M \rightarrow \infty$.

Since all curvature forms are closed, on the open part we have

$$\alpha^j - \beta^j = dd^c \left(\varphi \sum_{k=0}^{j-1} \alpha^k \wedge \beta^{j-1-k} \right).$$

Multiplying by $\chi_M \gamma$ and integrating by parts makes the difference of the integrals equal to

$$\int \varphi dd^c \chi_M \wedge \sum_{k=0}^{j-1} \alpha^k \wedge \beta^{j-1-k} \wedge \gamma.$$

Its absolute value is bounded on escaping tails by a constant times

$$\left(1 + \sum_i \log \rho_i\right) \Omega^n.$$

This is integrable: in a cusp variable the relevant radial integrals are bounded by $\int^\infty (1 + \log \rho) \rho^{-2} d\rho$. Thus the error tends to zero. The same domination permits removal of χ_M from the two curvature integrals, proving (9.6). Rational bundles are handled by first taking an integral tensor power and then dividing the forms and identities by that power.

Let r be the generic rank of the projective line map. Then $\alpha^{r+1} = 0$ on the open part, whereas α^r is nonzero and positive on a nonempty open subset. Apply (9.6) with complementary powers of an ample Kähler form. It gives vanishing of the $(r+1)$ st intersection power and strict positivity of the r th one. The nef intersection characterization proves (9.1). $\square$

9.2. Rank loss at a monodromy divisor.

Lemma 9.2 (Boundary rank drop by a Thom-form calculation). *In the setting of Lemma 9.1, suppose that monodromy is faithful on the moving highest line. If D_i is a boundary component with nonidentity transverse monodromy, then*

$$(9.7) \quad \nu(A|_{D_i}) < \nu(A).$$

Proof. Put $r = \nu(A)$. We first compute a bound for the numerical dimension on a boundary component of a unipotent model. This computation is unchanged on a component dominating D_i under a generically finite alteration, since restriction of the pulled-back line is the generically finite pullback of $A|_{D_i}$.

The limiting highest line. Near a general point of a unipotent boundary component write the coordinates as (t, s) , with divisor $t = 0$ and monodromy logarithm N . Let w be the weight and p the largest filtration index. The limiting mixed Hodge structure is polarized by N [25, Theorem 6.16]; its graded structures vary as polarized variations along the smooth boundary stratum [10, Proposition 2.10 and the following discussion]. The limiting F^p has dimension one and $F^{p+1} = 0$. Consequently it consists of one Deligne summand $I^{p,q}$. Write $p + q = w + l$. In the primitive decomposition for N , a nonzero N^k -image term with $k > 0$ would have to come from F^{p+k} , which is zero. The line is therefore primitive, $l \geq 0$, and, for a nonzero vector $u_0(s) = u(0, s)$ spanning it,

$$N^l u_0(s) \neq 0, \quad N^{l+1} u_0(s) = 0.$$

Shrink the boundary stratum so that its Hodge type, and hence l , is fixed.

For a highest Hodge vector in the pure variation, the Hodge norm is the flat polarization pairing with its conjugate, multiplied by the fixed factor determined by its Hodge type. Substituting the exponential frame into that pairing gives a polynomial in $y = -\log |t|$. Primitive polarization identifies its leading term as a positive constant times y^l times the squared norm $h_l(s)$ of the primitive graded line. Replacing $u(t, s)$ by $u_0(s)$ introduces errors $O(|t|)$ times powers of y ; the same holds after tangential differentiation, by holomorphicity of u . It follows, locally uniformly with tangential derivatives, that

$$(9.8) \quad \log \|v(t, s)\|^2 = l \log y + \log h_l(s) + c + O(y^{-1}) + O(|t|y^M)$$

for some M and a constant c . Thus the tangential curvature converges to the curvature of the primitive graded highest line. If q_i denotes the generic rank of its projective differential, the limiting tangential curvature has rank at most q_i .

The restriction intersection and crossing points. We claim that

$$(9.9) \quad \nu(A|_{D_i}) \leq q_i.$$

The following estimate proves the claim without assuming uniform convergence of (9.8) at crossings. Fix an ample Kähler form ω and an integer j with $q_i < j \leq n - 1$. Choose a smooth metric on $\mathcal{O}(D_i)$, write its curvature as β_i , and put $\tau = \log \|s_{D_i}\|^2$. Let g be a fixed smooth convex function with $g' = 0$ on $(-\infty, 0]$ and $g' = 1$ on $[\log 4, \infty)$, and set $f_\epsilon(x) = g(x - \log \epsilon^2)$. Then

$$(9.10) \quad \eta_\epsilon = \beta_i + dd^c f_\epsilon(\tau) = (1 - f'_\epsilon(\tau))\beta_i + \frac{i}{2\pi} f''_\epsilon(\tau) \partial\tau \wedge \bar{\partial}\tau$$

is a smooth closed representative of $c_1(\mathcal{O}(D_i))$. Here the normalization is $dd^c \log |t|^2 = [t = 0]$. The function $f_\epsilon(\tau)$ is constant near D_i , so it extends smoothly there. Outside $\|s_{D_i}\| \leq 2\epsilon$, the two curvature terms cancel and $\eta_\epsilon = 0$.

On a crossing chart put

$$\lambda = \frac{i dt \wedge d\bar{t}}{|t|^2}, \quad \rho = -\log |t|^2, \quad \Omega = \frac{\lambda}{\rho^2} + \Omega_s, \quad T = \alpha^j \wedge \omega^{n-1-j}.$$

The tangential form Ω_s is a product cusp metric, with Euclidean summands in the remaining coordinates. It has finite volume. The bound (9.2) gives $0 \leq T \leq C\Omega^{n-1}$. Write $\tau = \log |t|^2 + \psi$ with ψ smooth, and put $L_\epsilon = |\log \epsilon|$. The smooth first term of (9.10) contributes at most

$$C \int_{|t| \leq C\epsilon} \Omega^n = O(L_\epsilon^{-1}).$$

For the second term write $\partial\tau = a_0 + b_0$, with $a_0 = dt/t$ and $b_0 = \partial\psi$. Its support lies in $c\epsilon \leq |t| \leq C\epsilon$, with constants independent of s . On this shell, positivity gives

$$\begin{aligned} \int ia_0 \wedge \bar{a}_0 \wedge T &= O(1), \\ \int ib_0 \wedge \bar{b}_0 \wedge T &\leq C \int_{c\epsilon \leq |t| \leq C\epsilon} \Omega^n = O(L_\epsilon^{-2}). \end{aligned}$$

The first estimate follows because wedging with λ selects only the tangential part of T . For the second, a smooth positive form dominating $ib_0 \wedge \bar{b}_0$ is bounded by $C\Omega$. Cauchy–Schwarz for the positive form T now bounds the mixed terms by $O(L_\epsilon^{-1})$. The bounded coefficient f_ϵ'' does not change these estimates.

It remains to consider the pure normal-shell term. Set $t = \epsilon u$. Its cutoff coefficient becomes

$$g''(\log |u|^2 + \psi(\epsilon u, s)),$$

whose support is contained in a fixed annulus, since ψ is bounded. The absolute integrand is bounded, independently of ϵ , by

$$C \frac{i du \wedge d\bar{u}}{|u|^2} \wedge \Omega_s^{n-1}.$$

This is integrable on that annulus times the tangential chart, including all tangential divisor crossings. For almost every s , (9.8) gives a tangential curvature limit of rank less than j , so the tangential j th power tends to zero. Dominated convergence makes the pure normal-shell integral tend to zero. The estimates are uniform under truncating the tangential cusp variables, so no mass is lost when those truncations are removed. Choose a finite partition of unity subordinate to crossing charts. Multiplication by its bounded smooth functions preserves each absolute estimate. Summing the local estimates proves the same conclusion on all of D_i ; no integration by parts is performed after inserting this partition.

For each fixed ϵ , apply (9.6) with $\gamma = \omega^{n-1-j} \wedge \eta_\epsilon$. Its cohomology class is independent of ϵ . The preceding limit gives

$$c_1(A|_{D_i})^j \cdot [\omega|_{D_i}]^{n-1-j} = 0.$$

Nefness proves (9.9). If $q_i = n-1$, that inequality is automatic. This also explains why the generic limit calculation alone would not be sufficient: the uniform product bound is what controls the crossing locus.

A fixed quotient and transverse loops. Return to the original boundary component, with transverse monodromy $T_s \exp N$, where T_s has finite order. Work first in a transverse root coordinate killing T_s . Set $C_N = \text{coker } N$, endowed with the quotient of the fixed monodromy weight filtration. The primitive limiting line of weight $k = w + l$ projects nontrivially into $W_k C_N$. The fixed linear quotient

$$(9.11) \quad W_k C_N \longrightarrow \text{Gr}_k^W C_N$$

recovers its primitive graded line. Indeed, in the Lefschetz decomposition of Gr_k^W , quotienting by the image of N removes exactly the summands that are positive powers of N applied to higher primitives. The induced quotient is the primitive part of weight k . In particular, projective differential rank cannot increase under (9.11). No choice of a moving primitive splitting is involved.

The limiting line in C_N lies in one T_s -eigenspace. Compose the quotient by N with the fixed projection onto this eigenspace. Near the chosen general boundary point this gives a nonzero holomorphic projective map from the untwisted highest line. Under deck rotation the untwisted line is acted on by T_s ; on the projected eigenspace that action is scalar. The projective map therefore descends to the original chart. On the punctured chart it factors through the moving

highest-line map, since the exponential of N disappears in C_N . Its boundary differential rank is at least the rank q_i of the primitive graded line.

Suppose $\nu(A|_{D_i}) \geq r$. By (9.9), the projected map has at least r independent tangential differentials at a general boundary point. Choose r affine components $g_1, \dots, g_r$ with that property. Their differentials are independent along the boundary at that point, and the map $g = (g_1, \dots, g_r)$ extends holomorphically over the original chart. On the punctured chart it factors through the highest-line map. Lemma 9.1 gives the latter rank at most r . The hypotheses of the transverse-loop criterion (Lemma 8.6) are therefore satisfied in the original transverse coordinate, before the finite root cover. It follows that the original transverse monodromy fixes the moving line on the connected universal cover. This contradicts faithfulness and its nonidentity. When $r = 0$, use the same lemma with an empty list of functions; when $r > n - 1$, the supposed inequality $\nu(A|_{D_i}) \geq r$ is impossible on dimensional grounds. This proves Equation (9.7). $\square$

9.3. Connection with the integral projective argument. The hypotheses required for the preceding calculation arise from the integral tensor replacement and full-period descent in Lemmas 8.4 and 8.5. On the resulting projective base, Lemma 9.2 gives the same strict inequality as Lemma 8.7, with the boundary intersection justified directly by Thom forms. The boundary subtraction estimate of Lemma 8.8 then applies to the union J of the divisors with nonidentity transverse monodromy. The full-period map supplies the bigness of $K_R + J$ as in the proof of Theorem 8.1; the highest-line rank supplies the strict decrease in the degree of the restriction polynomials. Thus the two different period ranks retain their separate roles in adjoint positivity.

Remark 9.3 (Tensor replacement on a restricted locus). The same local results also give the curve-constancy part of the highest-line properties used in such comparisons. On a smooth complete curve, degree zero gives zero integral of the semipositive Hodge curvature by Equation (9.6), after an adapted cover if necessary. The curvature is therefore zero, and the highest line is projectively constant in a flat reference. If the highest line is projectively constant on a connected smooth algebraic locus, apply Lemma 8.4 to the restricted variation itself. Use its own monodromy image and isotypic decomposition: restricting an ambient replacement need not preserve faithfulness, because the restricted line image may have a larger projective stabilizer. Every monodromy element of the replacement fixes its constant highest line, so faithfulness makes that monodromy trivial. Since the new line is a positive tensor power of the original line, the original highest-line character on that locus has finite image. The tensor exponent here is allowed to depend on the restricted variation.

Remark 9.4. The adjoint conclusion is preserved under further smooth projective birational modification $\pi : R' \rightarrow R$, together with a compatible modification of the source. The parabolic pullback property preserves the line-bundle identity and nefness, while

$$K_{R'} + b\pi^*A = \pi^*(K_R + bA) + E_\pi, \quad E_\pi \geq 0,$$

preserves bigness. The calculations here concern the whole rank-one highest step of a polarizable integral pure variation on a projective pair; they assert no equality between highest-line rank, full-period rank, and whole-fiber birational variation.

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